ON BAYESIAN POSTERIOR MEAN ESTIMATORS IN IMAGING SCIENCES AND HAMILTON-JACOBI PARTIAL DIFFERENTIAL EQUATIONS

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ABSTRACT. Variational and Bayesian methods are two approaches that have been widely used to solve image reconstruction problems, but the question of which of these two approaches is preferable to apply for a given imaging problem remains contested. As variational methods correspond to maximum likelihood estimation in a Bayesian setting, this question is often centered around the merits and shortcomings of maximum a posteriori (MAP) estimators over Bayesian posterior mean (PM) estimators for image denoising problems. In this paper, we propose to describe a broad class of Bayesian methods and posterior mean estimators with Gaussian data fidelity term and log-concave prior using Hamilton–Jacobi (HJ) partial differential equations (PDEs). Where solutions to first-order HJ PDEs with initial data describe maximum a posteriori estimators in a Bayesian setting, here we show that solutions to some viscous HJ PDEs with initial data describe a broad class of posterior mean estimators. These connections allow us to establish several representation formulas and optimal estimate and variance inequalities. In particular, we use these connections to HJ PDEs to show that some posterior mean estimators can be expressed as proximal mappings of some smooth prior functions and derive the representation of these priors.

1. INTRODUCTION

Image denoising problems consist in estimating an unknown image from a noisy observation in a way that accounts for the underlying uncertainties. Variational and Bayesian methods have become two important approaches for doing so, but the question of which of these two approaches is preferable to apply for a given imaging problem remains contested [34, 35, 5, 23]. As variational methods correspond to maximum likelihood estimation in a Bayesian setting, this question is often centered around the merits and shortcomings of maximum a posteriori (MAP) estimators over Bayesian posterior mean (PM) estimators for image denoising problems. The goal of this paper is to describe a broad class of Bayesian posterior mean estimators (with Gaussian data fidelity term and log-concave prior) using Hamilton–Jacobi (HJ) partial differential equations (PDEs) and use

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these connections to clarify certain image denoising properties of this class of Bayesian posterior estimators.

To illustrate the main ideas of this paper, we first briefly introduce convex finite-dimensional variational methods relevant to image denoising problems and describe their connections to first-order HJ PDEs with initial data. We then discuss how these ideas can be extended to a broad class of Bayesian methods and posterior mean estimators.

Variational methods formulate image denoising problems as the optimization of a weighted sum of a data fidelity term (which models the noise corrupting the unknown image) and a prior (which models known properties of the unknown image) [7, 10], where the goal is to minimize this sum to obtain an estimate that hopefully accounts well for both the data fidelity term and the prior. This approach is popular because the resultant optimization problem for various non-smooth and convex prior functions used in image denoising problems, such as total variation and l_1-norm based priors, is well-understood [4, 6, 8, 10, 11, 15, 30] and can be solved efficiently using robust numerical optimization methods [7]. These methods have several shortcomings, however, in that reconstructed images from variational methods with non-smooth and convex priors often have undesirable and visually unpleasant staircasing effects due to the singularities of the non-smooth priors [38, 34, 53] and, somewhat paradoxically, may also substantially deviate from the prior and data fidelity term when these functions accurately model the prior knowledge about the unknown image and noise corrupting the image [34, 38, 46].

We will focus in this paper on the following class of variational imaging models: given an observed image $\boldsymbol{x} \in \mathbb{R}^n$, where $n \in \mathbb{N}$ corresponds to the number of pixels in the image, corrupted by additive Gaussian noise and a parameter $t > 0$, estimate the original uncorrupted image by solving the minimization problem

$$S_0(\boldsymbol{x}, t) := \inf_{\boldsymbol{y} \in \mathbb{R}^n} \left\{ \frac{1}{2t} \left\| \boldsymbol{x} - \boldsymbol{y} \right\|_2^2 + J(\boldsymbol{y}) \right\} \tag{1}$$

and computing its minimizer

$$\boldsymbol{u}_{MAP}(\boldsymbol{x}, t) := \arg\min_{\boldsymbol{y} \in \mathbb{R}^n} \left\{ \frac{1}{2t} \left\| \boldsymbol{x} - \boldsymbol{y} \right\|_2^2 + J(\boldsymbol{y}) \right\}. \tag{2}$$

The functions $\boldsymbol{y} \mapsto \frac{1}{2t} \left\| \boldsymbol{x} - \boldsymbol{y} \right\|_2^2$ and $J\colon \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ in (1) correspond, respectively, to the data fidelity term and prior of this class of variational models. The parameter $t > 0$ controls the relative importance of the data fidelity term over the prior in (1). When the prior J is proper, lower semicontinuous and convex (see Sect. 2 for the definitions), the solution $(\boldsymbol{x},t) \mapsto S(\boldsymbol{x},t)$ to the minimization problem (1) solves the first-order HJ equation with initial data (see Theorem 2.1 in Sect. 2, definition 9)

$$(3)\qquad \begin{cases} \dfrac{\partial S_0}{\partial t}(\boldsymbol{x},t) + \dfrac{1}{2}\left\|\nabla_{\boldsymbol{x}} S_0(\boldsymbol{x},t)\right\|_2^2 = 0 & \text{in } \mathbb{R}^n \times (0,+\infty), \\ S_0(\boldsymbol{x},0) = J(\boldsymbol{x}) & \text{in } \mathbb{R}^n. \end{cases}$$

The properties of the minimization problem (1) follow from the properties of the solution to this HJ equation [10]; the minimizer $\boldsymbol{u}_{MAP}(\boldsymbol{x},t)$, in particular, satisfies the representation formula

$$(4)\qquad \boldsymbol{u}(\boldsymbol{x},t) = \boldsymbol{x} - t\nabla_{\boldsymbol{x}} S_0(\boldsymbol{x},t).$$

The main idea of this paper is to interpret the minimization problem (1) in a Bayesian framework as the mode of an appropriate posterior distribution, use this distribution to generate the corresponding Bayesian posterior mean estimator, and finally relate the Bayesian posterior mean estimator to the solution to a viscous HJ PDE. Specifically, given a smoothing parameter $\epsilon > 0$, and under appropriate conditions on the prior J, we formally interpret the minimizer (2) as the mode or MAP estimate of the Bayesian posterior distribution

$$\boldsymbol{u}_{MAP}(\boldsymbol{x},t) = \operatorname*{arg\,min}_{\boldsymbol{y}\in\mathbb{R}^n} \left\{\frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + J(\boldsymbol{y})\right\} \equiv \operatorname*{arg\,max}_{\boldsymbol{y}\in\mathbb{R}^n} \left\{\frac{e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}}{\int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y}}\right\}.$$

The posterior distribution gives the Bayesian posterior mean estimate in terms of the observed image $\boldsymbol{x}$ and parameters t and ϵ:

$$(5)\qquad \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) := \frac{\int_{\mathbb{R}^n} \boldsymbol{y} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y}}{\int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y}}.$$

We will show that under appropriate conditions on the prior J, the posterior mean estimate (5) is described by the solution to a viscous HJ PDE with initial data J similarly to how the MAP estimate 2 is described by the solution to the first-order HJ PDE (22). Specifically, the function $S_\epsilon \colon \mathbb{R}^n \times [0,+\infty) \mapsto [0,+\infty)$ defined by

$$(6)\qquad S_\epsilon(\boldsymbol{x},t) := -\epsilon \ln\left(\frac{1}{(2\pi t\epsilon)^{n/2}} \int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y}\right)$$

solves the viscous HJ equation with initial data

$$\begin{cases} \frac{\partial S_\epsilon}{\partial t}(\boldsymbol{x},t) + \frac{1}{2}\left\|\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t)\right\|_2^2 = \frac{\epsilon}{2}\Delta_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t) & \text{in } \mathbb{R}^n \times (0,+\infty), \\ S_\epsilon(\boldsymbol{x},0) = J(\boldsymbol{x}) & \text{in } \mathbb{R}^n, \end{cases} \tag{7}$$

and the PM estimate satisfies the representation formula

$$\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) = \boldsymbol{x} - t\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t). \tag{8}$$

Various properties of PM estimators will follow from the properties to both the solutions $S_0(\boldsymbol{x},t)$ and $S_\epsilon(\boldsymbol{x},t)$ of these first-order and viscous HJ PDEs.

The main motivation for this paper comes from the papers of Louchet [34] and Gribonval [21] who showed that the class of Bayesian posterior mean estimators with Gaussian data fidelity term (5) and certain priors J can be expressed as minimizers to optimization problems involving a quadratic fidelity term and a smooth convex prior [34, 21, 22, 24, 23], i.e., there exists a smooth prior $f_{\text{prior}}\colon \mathbb{R}^n \mapsto \mathbb{R}$ such that

$$\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) = \inf_{\boldsymbol{y}\in\mathbb{R}^n}\left\{\frac{1}{2}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + f_{\text{prior(y)}}\right\}.$$

To our knowledge, Louchet [34] was the first to derive this result when the prior J in the posterior mean estimate (5) is a total variation prior [30]. This result was later generalized to a broader class of priors by Gribonval [21], and to colored Gaussian and other non-Gaussian data fidelity terms in [22, 23]. All these studies, however, only proved existence of such priors f_{prior}, however, and not their explicit form. The connections between HJ PDEs and posterior mean estimators that we derive in this paper will allow us to find the explicit form of this prior. [TODO: Explicit properties of the priors derived in these papers + mention image denoising properties (paper by Nikolova).]

In addition to its relevance to image denoising problems, the viscous HJ PDE (7) has recently received some attention in the deep learning literature, where its solution $\boldsymbol{x} \mapsto S_\epsilon(\boldsymbol{x},t)$ is known as the local entropy loss function and is a loss regularization effective at training deep networks [39, 9, 19, 50]. While this paper focuses on HJ PDEs and Bayesian posterior mean estimates in imaging sciences, the results in this paper should also be relevant to the deep learning literature and give new theoretical understandings of the local entropy loss function in terms of the data $\boldsymbol{x}$ and parameters t and ϵ.

Now to illustrate these properties, we give an example based on the Rudin–Osher–Fatemi (ROF) image denoising model, which consists of a total variation (TV) prior with quadratic data fidelity term [4, 8, 30]. The finite dimension anisotropic TV prior endowed with 4-nearest neighbors interactions [52], for instance, takes the form

$$TV(\boldsymbol{y}) = \frac{1}{2}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{N}(i)} |y_i - y_j|,$$

where the lattice $\mathcal{V}$ corresponds to a regular two-dimensional grid, and for every i we denote by $\mathcal{N}(i)$ the set of the 4-nearest neighbors of i. Letting the observed image $\boldsymbol{x}$ is defined on this lattice, with $|\mathcal{V}| = n$, the associated anisotropic ROF problem [30] then takes the form

$$\inf_{y_i\in\mathbb{R}:\, i\in\mathcal{V}} \left\{\sum_{i\in\mathcal{V}} \frac{1}{2t}(x_i - y_i)^2 + \frac{1}{2}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{N}(i)} |y_i - y_j|\right\} \equiv \inf_{\boldsymbol{y}\in\mathbb{R}^n} \left\{\frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + TV(\boldsymbol{y})\right\}. \tag{9}$$

In a Bayesian setting, the anisotropic ROF problem above corresponds to the mode of the Bayesian posterior distribution:

$$\boldsymbol{y} \mapsto \frac{e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + TV(\boldsymbol{y})\right)/\epsilon}}{\int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + TV(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y}} \tag{10}$$

and the corresponding posterior mean estimate is

$$\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) = \frac{\int_{\mathbb{R}^n} \boldsymbol{y} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + TV(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y}}{\int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + TV(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y}}.$$

Figure 1(A) depicts the standard Barbara picture (grayscale, 256x256), which we corrupt with Gaussian noise (zero mean with standard deviation $\sigma = 20$ with pixel values in $[0, 255]$) in Figure 1(B). The noisy image is denoised by computing the minimizer $\boldsymbol{u}_{MAP}(\boldsymbol{x}, t, \epsilon)$ to the ROF model 9 and illustrated in Figure 1(C), and the posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$ of the Bayesian posterior distribution 10 with $t = \epsilon = 20$ and illustrated in Figure 1. To compute the posterior mean estimate, we implemented the denoising algorithm outlined in Chapter 2 of [34] with convergence criterion outlined by Theorem 2.5 in Section 2.3 of [34]. [TODO: More details, including acceptance rate...].

1.1. **Contributions.** In this paper, we establish connections between a broad class of Bayesian methods and posterior mean estimators and solutions to HJ PDEs. These connections are described in Theorems 3.1 and 3.2 for viscous HJ PDEs and first-order HJ PDEs, respectively. We show in

FIGURE 1. The anisotropic ROF model for image denoising applied to the noisy grayscale 256x256 image of Barbara. The noise is Gaussian with zero mean and standard deviation $\sigma = 20$, and the parameters of the ROF model (9) and Bayesian posterior (10) are $t = 20$ and $\epsilon = 20$. (A) Original image of Barbare (B) Noisy image of Barbara (C) Denoised image of Barbara with the MAP estimate (D) Denoised image of Barbara with PM estimate.

Theorem 3.1 that the posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$ is described by the solution to the viscous HJ PDE (7) with initial data corresponding to the convex prior J, which we characterize in detail in terms of the data $\boldsymbol{x}$ and parameters t and ϵ. In particular, the posterior mean estimate satisfies the representation formula $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) = \boldsymbol{x} - t\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x}, t)$. Next, we use the connections between viscous HJ PDEs and posterior mean estimates established in Theorem 3.1 to show in Theorem 3.2 that the posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$ can be expressed through the gradient of the solution to a first-order HJ PDE with modified convex initial data $\boldsymbol{x} \mapsto K_\epsilon^*(\boldsymbol{x}, t) - \frac{1}{2}\|\boldsymbol{x}\|_2^2$, where

$$K_\epsilon(\boldsymbol{x}, t) = t\epsilon \ln\left(\frac{1}{(2\pi t\epsilon)^{n/2}} \int_{\text{dom } J} e^{\left(\frac{1}{t}\langle \boldsymbol{x}, \boldsymbol{y}\rangle - \frac{1}{2t}\|\boldsymbol{y}\|_2^2 - J(\boldsymbol{y})\right)/\epsilon} \, d\boldsymbol{y}\right)$$

and $\boldsymbol{x} \mapsto K_\epsilon^*(\boldsymbol{x}, t)$ is the Fenchel–Legendre transform of the function $\boldsymbol{x} \mapsto K_\epsilon(\boldsymbol{x}, t)$. In other words, we show

$$\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) = \underset{\boldsymbol{y}\in\mathbb{R}^n}{\arg\min} \left\{ \frac{1}{2} \|\boldsymbol{x} - \boldsymbol{y}\|_2^2 + \left(K_\epsilon^*(\boldsymbol{y}, t) - \frac{1}{2}\|\boldsymbol{y}\|_2^2 \right) \right\}.$$

This formula gives the representation of the convex prior enabling one to express the posterior mean estimate as the minimizer of a convex variational problem, and in fact in terms of the solution to a first-order HJ PDE, thereby extending the results of [34, 21] who showed existence of this prior when the data fidelity term is Gaussian, but not its representation. [TODO: Include discussion of smoothness if we derive infinite smoothness of the Fenchel transform of K?] [TODO: Mention again how connections to HJ PDEs are used.]

We also use the connections between posterior mean estimators and solutions to viscous HJ PDEs established in Theorem 3.1 to prove several topological, representation, and monotonicity properties of posterior mean estimators, respectively, in Propositions 4.1, 4.2, and 4.3. These properties are then used in Proposition4.4 to derive an optimal upper bound on the variance $\mathbb{E}_J \left[\|\boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)\|_2^2 \right]$, several estimates on the MAP and posterior mean estimates, and the behavior of the posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$ in the limit $t \to 0$. Finally, we use the connections between both MAP and posterior mean estimates and HJ PDEs to characterize the MAP and posterior mean estimates in terms of Bregman divergences. These results are presented in Theorem 4.1 and extend the findings of [5] and [40] by removing the restriction on the prior J to be uniformly Lipschitz continuous on $\mathbb{R}^n$, at least when the data fidelity term is Gaussian. This result does not generalize to the case where J is not defined everywhere on $\mathbb{R}^n$. Nonetheless, we show that in this situation there is a Bayes estimator, which we show can be described precisely in terms of the solution to both the first-order HJ PDE (2.1) and the viscous HJ PDE (3.1). [TODO: The previous paragraph is vague - fix after the body of the text is fixed.]

1.2. **Organization.** In Section 2, we review concepts of real and convex analysis that will be used throughout this paper. In Section 3, we establish theoretical connections between a broad class of Bayesian posterior mean estimators and HJ PDEs. Our mathematical set-up is described in Subsection 3.1, the connections of posterior mean estimators to viscous HJ PDEs are described in Subsection 3.2, and the connections of posterior mean estimators to first-order HJ PDEs are described in Subsection 3.3. We use these connections to establish various properties of posterior mean estimators in Section 4. Specifically, we establish topological, representation, and monotonicity properties of posterior mean estimators in Subsection 4.1, an optimal variance upper bound, various estimates, and the behavior of the posterior mean estimate (5) in the limit $t \to 0$ in Subsection 4.2. Finally, we establish properties of MAP and posterior mean estimators in terms of Bregman divergences in Subsection 4.3. Finally, we draw some conclusions and directions for future work in Sect. 5.

2. Background

This section reviews concepts from real and convex analysis that will be used in this paper. We refer the reader to [18, 26, 27, 44, 45] for comprehensive references. In what follows, the Euclidean scalar product on $\mathbb{R}^n$ will be denoted by $\langle \cdot, \cdot \rangle$ and its associated norm by $\|\cdot\|_2$. The closure and interior of a non-empty set $C \subset \mathbb{R}^n$ will be denoted by $\mathrm{cl}\ C$ and $\mathrm{int}\ C$, respectively. The boundary of a non-empty set $C \subset \mathbb{R}^n$ is defined as $\mathrm{cl}\ C \setminus \mathrm{int}\ C$ and will be denoted by $\mathrm{bd}\ C$. The domain of a function $f \colon \mathbb{R}^n \mapsto \mathbb{R} \cup \{+\infty\}$ is the set

$$\mathrm{dom}\ f = \{\boldsymbol{x} \in \mathbb{R}^n : f(\boldsymbol{x}) < +\infty\}.$$

We will denote the Borel σ-algebra on $\mathbb{R}^n$ by $\mathcal{B}(\mathbb{R}^n)$, and if given a Borel-measurable set $\Omega \subset \mathcal{B}(\mathbb{R}^n)$ and a Lebesgue measurable function $f \colon \mathbb{R}^n \mapsto \mathbb{R}$, we denote the Lebesgue integral of f over Ω by

$$\int_\Omega f(\boldsymbol{y})\, d\boldsymbol{y},$$

where $d\boldsymbol{y}$ denotes the n-dimensional Lebesgue measure.

Definition 1 (Proper and lower semicontinuous functions)**.** *A function $f \colon \mathbb{R}^n \mapsto \mathbb{R} \cup \{+\infty\}$ is proper if $\mathrm{dom}\ f \neq \varnothing$ and $f(\boldsymbol{x}) > -\infty$ for every $\boldsymbol{x} \in \mathrm{dom}\ f$.*

A function $f \colon \mathbb{R}^n \mapsto \mathbb{R} \cup \{+\infty\}$ is lower semicontinuous at $\boldsymbol{x} \in \mathbb{R}^n$ if it satisfies $\liminf_{k \to +\infty} f(\boldsymbol{x}_k) \geqslant f(\boldsymbol{x})$ for every sequence $\{\boldsymbol{x}_k\}_{k=1}^{+\infty} \subset \mathbb{R}^n$ such that $\lim_{k\to+\infty} \boldsymbol{x}_k = \boldsymbol{x}$.

Definition 2 (Convex sets and their relative interiors)**.** *A subset $C \subset \mathbb{R}^n$ is convex if for every pair $(\boldsymbol{x}, \boldsymbol{y}) \in C \times C$ and every scalar $\lambda \in (0, 1)$, the line segment $\lambda \boldsymbol{x} + (1 - \lambda)\boldsymbol{y} \in C$.*

The relative interior of a convex set C, denoted by ri (C)*, is the set of points in the interior of the unique smallest affine set containing C. Every convex set C with non-empty interior is n-dimensional with* $\mathrm{ri}\, C = \mathrm{int}\, C$ *and has positive Lebesgue measure, and furthermore the Lebesgue measure of the boundary* $\mathrm{bd}\, C$ *equals zero [31].*

Definition 3 (Convex functions and the set $\Gamma_0(\mathbb{R}^n)$)**.** *A proper function $f\colon \mathbb{R}^n \mapsto \mathbb{R} \cup \{+\infty\}$ is convex if its domain is convex and if for every pair $(\boldsymbol{x}, \boldsymbol{y}) \in \mathrm{dom}\, f \times \mathrm{dom}\, f$ and every scalar $\lambda \in (0,1)$, the inequality*

$$f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \leqslant \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}) \tag{11}$$

holds in $\mathbb{R} \cup \{+\infty\}$.

The class of proper, convex and lower semicontinuous functions is denoted by $\Gamma_0(\mathbb{R}^n)$.

A proper function f is strictly convex if the inequality is strict in (11) whenever $\boldsymbol{x} \neq \boldsymbol{y}$, and it is strongly convex with parameter $m > 0$ if for every pair $(\boldsymbol{x}, \boldsymbol{y}) \in \mathrm{dom}\, f \times \mathrm{dom}\, f$ and every scalar $\lambda \in (0,1)$, the inequality

$$f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \leqslant \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}) - \frac{m}{2}\lambda(1-\lambda) \left\|\boldsymbol{x} - \boldsymbol{y}\right\|_2^2.$$

holds in $\mathbb{R} \cup \{+\infty\}$.

A function $g\colon \mathbb{R}^n \mapsto (0, +\infty)$ is log-concave (respectively, strictly log-concave, strongly log-concave of parameter $m > 0$) if the function $-\ln(g)$ is convex (respectively strictly convex, strongly convex of parameter $m > 0$).

Definition 4 (Projections)**.** *Let C be a closed convex subset of $\mathbb{R}^n$. To every $\boldsymbol{x} \in \mathbb{R}^n$, there exists a unique element $\pi_C(\boldsymbol{x}) \in C$ called the projection of $\boldsymbol{x}$ onto C that is closest to $\boldsymbol{x}$ in Euclidean norm, i.e.,*

$$\pi_C(\boldsymbol{x}) \coloneqq \operatorname*{arg\,min}_{\boldsymbol{y} \in C} \left\|\boldsymbol{x} - \boldsymbol{y}\right\|_2^2. \tag{12}$$

This correspondence defines a map $\boldsymbol{x} \mapsto \pi_C(\boldsymbol{x})$ from $\mathbb{R}^n$ to C called the projector onto C ([2], Chapter 0.6, Corollary 1). It satisfies the following characterization:

$$\langle \boldsymbol{x} - \pi_C(\boldsymbol{x}), \boldsymbol{y} - \pi_C(\boldsymbol{x}) \rangle \leqslant 0, \quad \forall \boldsymbol{y} \in C. \tag{13}$$

Definition 5 (Subdifferentials and subgradients). *Let $f \in \Gamma_0(\mathbb{R}^n)$. The subdifferential of f at $\boldsymbol{x} \in \mathrm{dom}\ f$ is the set $\partial f(\boldsymbol{x})$ of vectors $\boldsymbol{p} \in \mathbb{R}^n$ that for every $\boldsymbol{y} \in \mathbb{R}^n$ satisfies the inequality*

$$f(\boldsymbol{y}) \geqslant f(\boldsymbol{x}) + \langle \boldsymbol{p}, \boldsymbol{y} - \boldsymbol{x} \rangle . \tag{14}$$

The vectors $\boldsymbol{p} \in \partial f(\boldsymbol{x})$ are called the subgradients of f at $\boldsymbol{x}$. The set of points $\boldsymbol{x} \in \mathrm{dom}\ f$ for which the subdifferential $\partial f(\boldsymbol{x})$ is non-empty is denoted by $\mathrm{dom}\ \partial f$. *and it includes the relative interior of the domain of f, that is,* $\mathrm{ri}\ (\mathrm{dom}\ f) \subset \mathrm{dom}\ \partial f$ *([44], Theorem 23.4).*

If f is strongly convex of parameter $m > 0$, then the subgradients of f at $\boldsymbol{x} \in \mathrm{dom}\ \partial f$ satisfy the stronger inequality

$$f(\boldsymbol{y}) \geqslant f(\boldsymbol{x}) + \langle \boldsymbol{p}, \boldsymbol{y} - \boldsymbol{x} \rangle + \frac{m}{2} \left\| \boldsymbol{y} - \boldsymbol{x} \right\|_2^2 .$$

If f is differentiable at $\boldsymbol{x}$, then $\boldsymbol{x} \in \mathrm{dom}\ \partial f$ and the gradient $\nabla f(\boldsymbol{x})$ is the unique subgradient of f at $\boldsymbol{x}$, and conversely if f has a unique subgradient at $\boldsymbol{x}$, then f is differentiable at that point ([44], Theorem 25.1).

The set-valued subdifferential mapping $\mathrm{dom}\ \partial f \ni \boldsymbol{x} \mapsto \partial f(\boldsymbol{x})$ *satisfies two important properties that will be used in this paper. First, it is monotone in that if f is strongly convex of parameter $m \geqslant 0$ for every pair $(\boldsymbol{x}_0, \boldsymbol{x}_1) \in \mathrm{dom}\ \partial f \times \mathrm{dom}\ \partial f$ and $\boldsymbol{p}_0 \in \partial f(\boldsymbol{x}_0)$, $\boldsymbol{p}_1 \in \partial f(\boldsymbol{x}_1)$, the following inequality holds ([44], page 240 and Corollary 31.5.2):*

$$\langle \boldsymbol{p}_0 - \boldsymbol{p}_1, \boldsymbol{x}_0 - \boldsymbol{x}_1 \rangle \geqslant m \left\| \boldsymbol{x}_0 - \boldsymbol{x}_1 \right\|_2^2 . \tag{15}$$

Second, for every $\boldsymbol{x} \in \mathrm{dom}\ \partial f$ the subdifferential $\partial f(\boldsymbol{x})$ is a closed convex set, and as a consequence, the mapping $\mathrm{dom}\ \partial f \ni \boldsymbol{x} \mapsto \pi_{\partial f(\boldsymbol{x})}(\boldsymbol{0})$ *selects the subgradient of the minimal norm in $\partial f(\boldsymbol{x})$ and defines a function continuous almost everywhere on* $\mathrm{dom}\ \partial f$, *which is a consequence of the fact that this mapping agrees with the gradient of f over the set of points in* $\mathrm{int}\ \mathrm{dom}\ J$ *at which f is differentiable ([44], Theorem 25.5).*

Definition 6 (Fenchel–Legendre transform). *Let $f \in \Gamma_0(\mathbb{R}^n)$. The Fenchel–Legendre transform $f^* \colon \mathbb{R}^n \mapsto \mathbb{R} \cup \{+\infty\}$ of f is defined by*

$$f^*(\boldsymbol{p}) = \sup_{\boldsymbol{x} \in \mathbb{R}^n} \left\{ \langle \boldsymbol{p}, \boldsymbol{x} \rangle - f(\boldsymbol{x}) \right\} . \tag{16}$$

For every $f \in \Gamma_0(\mathbb{R}^n)$, the mapping $f \mapsto f^*$ is one-to-one, $f^* \in \Gamma_0(\mathbb{R}^n)$, and $(f^*)^* = f$. Moreover, for every $\boldsymbol{x} \in \mathbb{R}^n$ and $\boldsymbol{p} \in \mathbb{R}^n$, f and f^* satisfy Fenchel's inequality

$$f(\boldsymbol{x}) + f(\boldsymbol{p}) \geqslant \langle \boldsymbol{x}, \boldsymbol{p} \rangle, \tag{17}$$

where quality holds if and only if $\boldsymbol{p} \in \partial f(\boldsymbol{x})$, if and only if $\boldsymbol{x} \in \partial f^*(\boldsymbol{p})$ ([27], corollary 1.4.4). If f is also differentiable, the supremum in (16) is attained whenever there exists $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{p} = \nabla f(\boldsymbol{x})$.

Definition 7 (Infimal convolutions)**.** *Let $f_1 \in \Gamma_0(\mathbb{R}^n)$ and $f_2 \in \Gamma_0(\mathbb{R}^n)$. The infimal convolution of f_1 and f_2 is the function*

$$\mathbb{R}^n \ni \boldsymbol{x} \mapsto (f_1 \square f_2)(\boldsymbol{x}) = \inf_{\boldsymbol{x}_1 + \boldsymbol{x}_2 = \boldsymbol{x}} \left\{ f_1(\boldsymbol{x}_1) + f_2(\boldsymbol{x}_2) \right\}. \tag{18}$$

The infimal convolution is exact if the infimum is attained at $\boldsymbol{x}_1 \in \mathrm{dom}\ f_1$ and $\boldsymbol{x}_2 \in \mathrm{dom}\ f_2$, and in that case the infimum in (18) *can be replaced by a minimum. The Fenchel–Legendre transform of the infimal convolution (18) is the sum of their respective Fenchel–Legendre transforms ([44], Theorem 16.4), that is,*

$$(f_1 \square f_2)^* (\boldsymbol{p}) = f_1^*(\boldsymbol{p}) + f_2^*(\boldsymbol{p}).$$

If $f \in \Gamma_0(\mathbb{R}^n)$, then Moreau's Theorem [37, 28] asserts that

$$\frac{1}{2} \left\| \cdot \right\|_2^2 \square f + \frac{1}{2} \left\| \cdot \right\|_2^2 \square f^* = \frac{1}{2} \left\| \cdot \right\|_2^2,$$

and conversely, if g and h are two convex functions on $\mathbb{R}^n$ such that $g + h = \frac{1}{2} \left\| \cdot \right\|_2^2$, then there exists a unique function $F \in \Gamma_0(\mathbb{R}^n)$ such that

$$g = \frac{1}{2} \left\| \cdot \right\|_2^2 \square F, \text{ and } h = \frac{1}{2} \left\| \cdot \right\|_2^2 \square F^*,$$

where $F(\boldsymbol{x}) = h^*(\boldsymbol{x}) - \frac{1}{2} \left\| \boldsymbol{x} \right\|_2^2$ for every $\boldsymbol{x} \in \mathbb{R}^n$, and moreover g and f are continuously differentiable and

$$\nabla g(\boldsymbol{x}) \in \partial F(\nabla h(\boldsymbol{x})) \quad \text{and} \quad \nabla h(\boldsymbol{x}) \in \partial F^*(\nabla g(\boldsymbol{x})).$$

Definition 8 (Bregman divergences)**.** *Let $f \in \Gamma_0(\mathbb{R}^n)$. The Bregman divergence of f is the function $D_f \colon \mathbb{R}^n \times \mathbb{R}^n \mapsto \mathbb{R} \cup \{+\infty\}$ defined by*

$$D_f(\boldsymbol{x}, \boldsymbol{p}) = f(\boldsymbol{x}) - \langle \boldsymbol{p}, \boldsymbol{x} \rangle + f^*(\boldsymbol{p}). \tag{19}$$

It satisfies $D_f(\boldsymbol{x}, \boldsymbol{p}) \geqslant 0$ *for every* $\boldsymbol{x} \in \mathbb{R}^n$ *and* $\boldsymbol{p} \in \mathbb{R}^n$ *by Fenchel's inequality (17), with* $D_f(\boldsymbol{x}, \boldsymbol{p}) = 0$ *whenever* $\boldsymbol{p} \in \partial f(\boldsymbol{x})$. *It also satisfies* $D_f(\boldsymbol{x}, \boldsymbol{p}) = D_{f^*}(\boldsymbol{p}, \boldsymbol{x})$, *with* $D_f(\boldsymbol{x}, \boldsymbol{p}) = D_f(\boldsymbol{p}, \boldsymbol{x})$ *if and only if* f *is the quadratic* $f = \frac{1}{2}\left\|\cdot\right\|_2^2$.

Definition 9 (Moreau–Yosida envelopes and proximal mappings)**.** *Let* $t > 0$, *and* $J \in \Gamma_0(\mathbb{R}^n)$. *The functions*

$$\boldsymbol{x} \mapsto \inf_{\boldsymbol{y} \in \mathbb{R}^n} \left\{ \frac{1}{2t} \left\|\boldsymbol{x} - \boldsymbol{y}\right\|_2^2 + J(\boldsymbol{y}) \right\} \tag{20}$$

and

$$\boldsymbol{x} \mapsto \operatorname*{arg\,min}_{\boldsymbol{y} \in \mathbb{R}^n} \left\{ \frac{1}{2t} \left\|\boldsymbol{x} - \boldsymbol{y}\right\|_2^2 + J(\boldsymbol{y}) \right\} \tag{21}$$

are called the Moreau–Yosida envelope and proximal mapping of J, *respectively [27, 37, 45]. Their properties have been extensively studied in convex and functional analysis, and they form the basis for the mathematical analysis of the convex variational imaging model* (1) *and corresponding minimizer* (2) *[10]. The following theorem describes the behavior of both the solution to the infimum problem* (1) *and its corresponding minimizer* (2), *and it shows in particular that for any observed image* $\boldsymbol{x} \in \mathbb{R}^n$ *and parameter* $t > 0$, *the imaging problem* (1) *has always a unique solution. The readers may refer to Darbon [10] for more details.*

Theorem 2.1. *Suppose* J *is a proper, lower semicontinuous and convex function, i.e.,* $J \in \Gamma_0(\mathbb{R}^n)$. *Then the following statements hold.*

(i) *The unique continuously differentiable and convex function* $S_0 \colon \mathbb{R}^n \times [0, +\infty) \mapsto \mathbb{R}$ *that satisfies the first-order Hamilton–Jacobi equation with initial data*

$$\begin{cases} \dfrac{\partial S_0}{\partial t}(\boldsymbol{x}, t) + \dfrac{1}{2}\left\|\nabla_{\boldsymbol{x}} S_0(\boldsymbol{x}, t)\right\|_2^2 = 0 & \text{in } \mathbb{R}^n \times (0, +\infty), \\ S_0(\boldsymbol{x}, 0) = J(\boldsymbol{x}) & \text{in } \mathbb{R}^n, \end{cases} \tag{22}$$

is defined by

$$S_0(\boldsymbol{x}, t) = \left(\frac{t}{2}\left\|\cdot\right\|_2^2 + J^* \right)^* (\boldsymbol{x}) \qquad \text{(Hopf formula)} \tag{23}$$

$$= \left((\frac{t}{2}\left\|\cdot\right\|_2^2)^* \square J \right)(\boldsymbol{x}) \qquad \text{(Lax–Oleinik formula)} \tag{24}$$

$$= \inf_{\boldsymbol{y} \in \mathbb{R}^n} \left\{ \frac{1}{2t}\left\|\boldsymbol{x} - \boldsymbol{y}\right\|_2^2 + J(\boldsymbol{y}) \right\}. \tag{25}$$

Furthermore, for every $\boldsymbol{x} \in \mathrm{dom}\ J$, sequence $\{t_k\}_{k=1}^{+\infty}$ of positive real numbers converging to 0, and sequence $\{\boldsymbol{d}_k\}_{k=1}^{+\infty}$ of vectors converging to $\boldsymbol{d} \in \mathbb{R}^n$, the pointwise limit $S_0(\boldsymbol{x}+t_k\boldsymbol{d}_k, t_k)$ as $k \to +\infty$ exists and satisfies

$$\lim_{k\to+\infty} S_0(\boldsymbol{x} + t_k\boldsymbol{d}_k, t_k) = J(\boldsymbol{x}).$$

(ii) For every $\boldsymbol{x} \in \mathbb{R}^n$ and $t > 0$, the infimum in (25) exists and is attained at a unique point $\boldsymbol{u}_{MAP}(\boldsymbol{x}, t) \in \mathrm{dom}\ \partial J$. In addition, $\nabla_{\boldsymbol{x}} S_0(\boldsymbol{x}, t) = \left(\frac{\boldsymbol{x}-\boldsymbol{u}_{MAP}(\boldsymbol{x},t)}{t}\right) \in \partial J(\boldsymbol{u}_{MAP}(\boldsymbol{x}, t))$, and the minimizer $\boldsymbol{u}_{MAP}(\boldsymbol{x}, t)$ satisfies the formula

$$\boldsymbol{u}_{MAP}(\boldsymbol{x}, t) = \boldsymbol{x} - t\nabla_{\boldsymbol{x}} S_0(\boldsymbol{x}, t). \tag{26}$$

(iii) For every $\boldsymbol{x} \in \mathrm{dom}\ J$, the pointwise limit of $\boldsymbol{u}_{MAP}(\boldsymbol{x}, t)$ as $t \to 0$ exists and satisfies

$$\lim_{\substack{t\to 0\\ t>0}} \boldsymbol{u}_{MAP}(\boldsymbol{x}, t) = \boldsymbol{x}.$$

(iv) Let $\boldsymbol{x} \in \mathrm{dom}\ \partial J$ and let $\{t_k\}_{k=1}^{+\infty}$ be a sequence of positive real numbers converging to zero. Then the limit of $\nabla_{\boldsymbol{x}} S_0(\boldsymbol{x}, t_k)$ as $k \to +\infty$ exists and satisfies

$$\lim_{k\to+\infty} \nabla_{\boldsymbol{x}} S_0(\boldsymbol{x}, t_k) = \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0}). \tag{27}$$

Proof. The proof of (i) relies on convex analysis; see Hiriart-Urruty [25] for a detailed proof. A proof of (ii)-(iv) can be found in Darbon [10] (Lemma 2.1, Proposition 3.1, Proposition 3.2 (i), and Proposition 3.4). □

3. Connections between Bayesian posterior mean estimators and Hamilton–Jacobi partial differential equations

3.1. Set-up. To establish connections between Bayesian posterior mean estimators and Hamilton–Jacobi equations, we will assume that the imaging prior J in the variational imaging model (1) satisfies the following assumptions:

(A1) $J \in \Gamma_0(\mathbb{R}^n)$,

(A2) $\mathrm{int}\ (\mathrm{dom}\ J) \neq \varnothing$,

(A3) $\inf_{\boldsymbol{y}\in\mathbb{R}^n} J(\boldsymbol{y}) < +\infty$, and without loss of generality, $\inf_{\boldsymbol{y}\in\mathbb{R}^n} J(\boldsymbol{y}) = 0$.

Assumption (A1) ensures that the minimal value of the convex imaging problem (1) and its minimizer (2) are well-defined and enjoy several properties (see Theorem 2.1). Assumption (A2) ensures that for every $\boldsymbol{x} \in \mathbb{R}^n$, $t > 0$, and $\epsilon > 0$, the posterior distribution

$$\mathbb{R}^n \times \mathbb{R}^n \times (0, +\infty) \times (0, +\infty) \ni (\boldsymbol{y}, \boldsymbol{x}, t, \epsilon) \mapsto q(\boldsymbol{y}|\boldsymbol{x}, t, \epsilon) = \frac{e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}}{\int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon} \, d\boldsymbol{y}} \tag{28}$$

and its associated partition function

$$\mathbb{R}^n \times (0, +\infty) \times (0, +\infty) \ni (\boldsymbol{x}, t, \epsilon) \mapsto Z_J(\boldsymbol{x}, t, \epsilon) = \int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon} \, d\boldsymbol{y} \tag{29}$$

are well-defined, and finally assumption (A3) guarantees that the partition function is also bounded from above independently of $\boldsymbol{x} \in \mathbb{R}^n$. Additional requirements beyond that $J \in \Gamma_0(\mathbb{R}^n)$ are necessary because the integral in (29) adds a measure-theoretic aspect largely absent from the convex minimization problem (1). For convenience, in the rest of this paper we will denote the expected value of a measurable function $f\colon (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n)) \mapsto (\mathbb{R}^m, \mathcal{B}(\mathbb{R}^m))$ (with $m \in \mathbb{N}$) by

$$\mathbb{E}_J\left[f(\boldsymbol{y})\right] = \frac{1}{Z_J(\boldsymbol{x}, t, \epsilon)} \int_{\mathbb{R}^n} f(\boldsymbol{y}) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon} \, d\boldsymbol{y}. \tag{30}$$

The posterior mean estimate and variance of the posterior distribution (28) can be expressed as $\mathbb{E}_J\left[\boldsymbol{y}\right]$ and $\mathbb{E}_J\left[\|\boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)\|_2^2\right]$, respectively.

3.2. **Connections to second-order Hamilton–Jacobi equations.** The next theorem establishes connections between viscous HJ PDEs with initial data J satisfying assumptions (A1)-(A3) and both the partition function (29) and the Bayesian posterior mean estimate (5). These connections mirror those between the first-order HJ PDE (22) with initial data J satisfying assumption (A1) and both the convex minimization problem (1) and the MAP estimate (2). The connections between viscous HJ PDEs and Bayesian posterior mean estimators will be leveraged later to describe various properties of posterior mean estimators in terms of the observed image $\boldsymbol{x}$ and parameters t and ϵ, and in particular in Section 3.3 to show that the posterior mean estimate (5) can be expressed as the minimizer associated to the solution to a first-order HJ PDE (Theorem 3.2).

Theorem 3.1 (The viscous Hamilton–Jacobi equation with initial data in $\Gamma_0(\mathbb{R}^n)$)**.** *Suppose the function J satisfies assumptions (A1)-(A3). Then the following statements hold.*

(i) For every $\epsilon > 0$*, the function* $S_\epsilon \colon \mathbb{R}^n \times [0,+\infty) \mapsto [0,+\infty)$ *defined by*

$$S_\epsilon(\boldsymbol{x},t) \coloneqq -\epsilon \ln\left(\frac{1}{(2\pi t\epsilon)^{n/2}} Z_J(\boldsymbol{x},t,\epsilon)\right) = -\epsilon \ln\left(\frac{1}{(2\pi t\epsilon)^{n/2}} \int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y}\right) \tag{31}$$

is the unique smooth solution to the second-order Hamilton Jacobi PDE with initial data

$$\begin{cases} \dfrac{\partial S_\epsilon}{\partial t}(\boldsymbol{x},t) + \dfrac{1}{2}\left\|\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t)\right\|_2^2 = \dfrac{\epsilon}{2}\Delta_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t) & \text{in } \mathbb{R}^n \times (0,+\infty), \\ S_\epsilon(\boldsymbol{x},0) = J(\boldsymbol{x}) & \text{in } \mathbb{R}^n. \end{cases} \tag{32}$$

In addition, the domain of integration in (3.1) *can be taken to be* dom J *or, up to a set of Lebesgue measure zero,* int (dom J) *or* dom (∂J)*. Furthermore, for every* $\boldsymbol{x} \in \operatorname{dom} J$ *and* $\epsilon > 0$*, except possibly at the boundary points* $\boldsymbol{x} \in (\operatorname{dom} J)\backslash(\operatorname{int}(\operatorname{dom} J))$ *if such points exist, the pointwise limit* $S_\epsilon(\boldsymbol{x},t)$ *as* $t \to 0$ *exists and satisfies*

$$\lim_{\substack{t\to 0\\ t>0}} S_\epsilon(\boldsymbol{x},t) = e^{-J(\boldsymbol{x})/\epsilon}. \tag{33}$$

If $\boldsymbol{x} \in (\operatorname{dom} J)\backslash(\operatorname{int}(\operatorname{dom} J))$*, then we may only conclude*

$$\liminf_{\substack{t\to 0\\ t>0}} S_\epsilon(\boldsymbol{x},t) \geqslant J(\boldsymbol{x})$$

and

$$\limsup_{\substack{t\to 0\\ t>0}} S_\epsilon(\boldsymbol{x},t) \leqslant J(\boldsymbol{x}) - \epsilon \ln\left(\frac{1}{(2\pi\epsilon)^{n/2}} \int_{\operatorname{dom}\ J} e^{-\frac{1}{2\epsilon}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2}\, d\boldsymbol{y}\right).$$

(ii) (Convexity and monotonicity properties).

(a) The function $\mathbb{R}^n \times (0,+\infty) \ni (\boldsymbol{x},t) \mapsto S_\epsilon(\boldsymbol{x},t) - \frac{n\epsilon}{2}\ln t$ *is jointly convex.*

(b) The function $(0,+\infty) \ni t \mapsto S_\epsilon(\boldsymbol{x},t) - \frac{n\epsilon}{2}\ln t$ *is strictly monotone decreasing.*

(c) The function $(0,+\infty) \ni \epsilon \mapsto S_\epsilon(\boldsymbol{x},t) - \frac{n\epsilon}{2}\ln \epsilon$ *is strictly monotone decreasing.*

(d) The function $\mathbb{R}^n \ni \boldsymbol{x} \mapsto \frac{1}{2}\|\boldsymbol{x}\|_2^2 - tS_\epsilon(\boldsymbol{x},t)$ *is convex.*

(iii) (Connections to the posterior mean and variance) The posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)$ *and the variance* $\mathbb{E}_J\left[\|\boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)\|_2^2\right]$ *satisfy the formulas*

$$\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) = \boldsymbol{x} - t\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t) \tag{34}$$

and

$$\mathbb{E}_J\left[\|\boldsymbol{y}-\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)\|_2^2\right] = t\epsilon\nabla_{\boldsymbol{x}}\cdot\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) = nt\epsilon - t^2\epsilon\Delta_{\boldsymbol{x}}S_\epsilon(\boldsymbol{x},t). \tag{35}$$

(iv) *Let $S_0 : \mathbb{R}^n \times (0,+\infty) \mapsto \mathbb{R}$ denote the continuously differentiable and convex solution to the first-order HJ PDE* (22) *with initial data J. For every $\boldsymbol{x} \in \mathbb{R}^n$ and $t > 0$, the following limit holds:*

$$\lim_{\substack{\epsilon\to 0\\ \epsilon>0}} -\epsilon\ln\left(\frac{1}{(2\pi t\epsilon)^{n/2}}\int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon}\,d\boldsymbol{y}\right) = \inf_{\boldsymbol{y}\in\mathbb{R}^n}\left\{\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right\}, \tag{36}$$

that is,

$$\lim_{\substack{\epsilon\to 0\\ \epsilon>0}} S_\epsilon(\boldsymbol{x},t) = S_0(\boldsymbol{x},t),$$

and the limit converges uniformly over every closed bounded set of $\mathbb{R}^n \times (0,+\infty)$ in $(\boldsymbol{x},t)$. In addition, the gradient $\nabla_{\boldsymbol{x}}S_\epsilon(\boldsymbol{x},t)$, the partial derivative $\frac{\partial S_\epsilon(\boldsymbol{x},t)}{\partial t}$, and the Laplacian $\frac{\epsilon}{2}\Delta_{\boldsymbol{x}}S_\epsilon(\boldsymbol{x},t)$ satisfy the limits

$$\lim_{\substack{\epsilon\to 0\\ \epsilon>0}} \nabla_{\boldsymbol{x}}S_\epsilon(\boldsymbol{x},t) = \nabla_{\boldsymbol{x}}S_0(\boldsymbol{x},t), \quad \lim_{\substack{\epsilon\to 0\\ \epsilon>0}} \frac{\partial S_\epsilon}{\partial t}(\boldsymbol{x},t) = \frac{\partial S_0}{\partial t}(\boldsymbol{x},t),$$

and

$$\lim_{\substack{\epsilon\to 0\\ \epsilon>0}} \frac{\epsilon}{2}\Delta_{\boldsymbol{x}}S_\epsilon(\boldsymbol{x},t) = 0,$$

where each limit converges uniformly over every closed bounded set of $\mathbb{R}^n \times (0,+\infty)$ in $(\boldsymbol{x},t)$. As a consequence, for every $\boldsymbol{x} \in \mathbb{R}^n$ and $t > 0$, the pointwise limit of $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)$ as $\epsilon \to 0$ exists and satisfy

$$\lim_{\substack{\epsilon\to 0\\ \epsilon>0}} \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) = \boldsymbol{u}_{MAP}(\boldsymbol{x},t),$$

and the limit converges uniformly over every closed bounded set of $\mathbb{R}^n \times (0,+\infty)$ in $(\boldsymbol{x},t)$.

Proof. See Appendix A. □

Remark 3.1. *Note that solutions to* (3.2) *exist under weaker conditions by weakening assumptions (A1) and (A3), but then global existence, pointwise limit to the initial condition (almost everywhere), boundedness, and log–concavity properties of solutions may no longer hold.*

To illustrate certain aspects of Theorem 3.1 and properties of posterior mean estimates, we give here two analytical examples.

Example 1 (Tikhonov–Phillips regularization)**.** *Let* $J(\boldsymbol{x}) = \frac{m}{2}\|\boldsymbol{x}\|_2^2$ *with* $m > 0$*, and consider the solution* $S_0(\boldsymbol{x}, t)$ *and* $S_\epsilon(\boldsymbol{x}, t)$ *to the first-order PDE* (22) *and viscous HJ PDE* (32) *with initial data* J*, respectively.*

The solution $S_0(\boldsymbol{x}, t)$ *is given by the Lax–Oleinik formula*

$$\begin{aligned} S_0(\boldsymbol{x}, t) &= \inf_{\boldsymbol{y}\in\mathbb{R}^n}\left\{\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + \frac{m}{2}\|\boldsymbol{y}\|_2^2\right\} \\ &= \frac{m\|\boldsymbol{x}\|_2^2}{2(1+mt)}. \end{aligned}$$

This minimization problem is a special case of Tikhonov–Phillips regularization (also known as ridge regression in statistics), a method for regularizing ill-posed problems in inverse problems and statistics using a quadratic prior [42, 48, 47]. The corresponding minimizer can be computed using the gradient $\nabla_{\boldsymbol{x}} S_0(\boldsymbol{x}, t)$ *via equation* (26) *in Theorem 3.1:*

$$\boldsymbol{u}_{MAP}(\boldsymbol{x}, t) = \boldsymbol{x} - t\nabla_{\boldsymbol{x}} S_0(\boldsymbol{x}, t) = \boldsymbol{x} - \frac{mt\boldsymbol{x}}{1+mt} = \frac{\boldsymbol{x}}{1+mt}.$$

The solution $S_\epsilon(\boldsymbol{x}, t)$ *is given by the integral*

$$\begin{aligned} S_\epsilon(\boldsymbol{x}, t) &= -\epsilon \ln\left(\frac{1}{(2\pi t\epsilon)^{n/2}}\int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + \frac{m}{2}\|\boldsymbol{y}\|_2^2\right)}\, d\boldsymbol{y}\right) \\ &= \frac{m\|\boldsymbol{x}\|^2}{2(1+mt)} + \frac{n\epsilon}{2}\ln(1+mt). \end{aligned}$$

The posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$ *can be computed using the representation formula* (34) *in Theorem 3.1(iii) by calculating the gradient* $\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x}, t)$*:*

$$\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) = \boldsymbol{x} - t\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x}, t) = \boldsymbol{x} - \frac{mt\boldsymbol{x}}{1+mt} = \frac{\boldsymbol{x}}{1+mt}.$$

The variance $\mathbb{E}_J\left[\|\boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)\|_2^2\right]$ *can be computed using the representation formula* (35) *in Theorem 3.1(iii) by calculating the divergence of* $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$*:*

$$\mathbb{E}_J\left[\|\boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)\|_2^2\right] = t\epsilon\nabla_{\boldsymbol{x}} \cdot \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) = \frac{nt\epsilon}{1+mt}. \tag{37}$$

Comparing the solutions $S_0(\boldsymbol{x},t)$ and $S_\epsilon(\boldsymbol{x},t)$, we see that $\lim_{\substack{\epsilon\to 0\\ \epsilon>0}} S_\epsilon(\boldsymbol{x},t) = S_0(\boldsymbol{x},t)$ for every $\boldsymbol{x} \in \mathbb{R}^n$ and $t > 0$, in accordance to the result established in Theorem 3.1(iv). Note also that while $(\boldsymbol{x},t) \mapsto S_0(\boldsymbol{x},t)$ is jointly convex, its viscous counterpart $(\boldsymbol{x},t) \mapsto S_\epsilon(\boldsymbol{x},t)$ is not. Indeed, $t \mapsto S_\epsilon(\boldsymbol{x},t)$ is not convex. It is convex only after subtracting $\frac{n\epsilon}{2}\ln t$ from $S_\epsilon(\boldsymbol{x},t)$. This implies that the joint convexity result Theorem 3.1(ii)(a) cannot be improved, at least without further assumptions on the initial data J.

Example 2 (Soft thresholding)**.** *Let $J(\boldsymbol{x}) = \sum_{i=1}^n \lambda_i |\boldsymbol{x}_i|$, where $\lambda_i > 0$ for each $i \in \{1,\dots,n\}$, and consider the solutions $S_0(\boldsymbol{x},t)$ and $S_\epsilon(\boldsymbol{x},t)$ to the first-order PDE* (22) *and second-order PDE* (32) *with initial data J, respectively.*

The solution $S_0(\boldsymbol{x},t)$ is given by the Lax–Oleinik formula

$$
\begin{aligned}
S_0(\boldsymbol{x},t) &= \inf_{\boldsymbol{y}\in\mathbb{R}^n} \left\{ \frac{1}{2t} \left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + \sum_{i=1}^n \lambda_i |y_i| \right\} \\
&= \sum_{i=1}^n \left(\inf_{y_i\in\mathbb{R}} \left\{ \frac{1}{2t}(x_i-y_i)^2 + \lambda_i |y_i| \right\} \right),
\end{aligned}
$$

where x_i and y_i denote the i^{th} component of the vectors $\boldsymbol{x}$ and $\boldsymbol{y}$, respectively. In the context of imaging, this minimization problem corresponds to denoising an image with the weighted sum of a quadratic fidelity term and a weighted l_1-norm as the prior. This prior is widely used in imaging to encourage sparsity of an image, and it has received considerable interest due to its connection with compressed sensing reconstruction [6, 15]. The solution to this minimization problem corresponds to a soft thresholding applied component-wise to the vector $\boldsymbol{x}$ [12, 17, 33]. The soft thresholding operator is defined for any real number a and positive real number α as

$$
\mathbb{R}\times(0,+\infty) \ni (a,\alpha) \mapsto T(a,\alpha) = \begin{cases} a-\alpha & \text{if } a>\alpha, \\ 0 & \text{if } a\in[-\alpha,\alpha], \\ a+\alpha & \text{if } a<-\alpha. \end{cases} \tag{38}
$$

The minimizer in the Lax–Oleinik formula of $S_0(\boldsymbol{x},t)$ is then given component-wise by

$$
\left(\boldsymbol{u}_{MAP}(\boldsymbol{x},t)\right)_i = T(x_i, t\lambda_i),
$$

so that

$$S_0(\boldsymbol{x},t) = \sum_{i=1}^{n} \left(\frac{1}{2t}(x_i - T(x_i, t\lambda_i))^2 + \lambda_i \left| T(x_i, t\lambda_i) \right| \right).$$

The solution $S_\epsilon(\boldsymbol{x},t)$ *is given by the integral*

$$\begin{aligned} S_\epsilon(\boldsymbol{x},t) &= -\epsilon \ln \left(\frac{1}{(2\pi t\epsilon)^{n/2}} \int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + \sum_{k=1}^{n} \lambda_i |y_i|\right)/\epsilon} \, d\boldsymbol{y} \right) \\ &= -\epsilon \sum_{i=1}^{n} \ln \left(\frac{1}{2}\sqrt{\frac{2}{\pi t\epsilon}} \int_{-\infty}^{+\infty} e^{-\left(\frac{1}{2t}(x_i - y_i)^2 + \lambda_i |y_i|\right)/\epsilon} \, dy_i \right) \\ &= -\epsilon \sum_{i=1}^{n} \ln \left(\frac{1}{2}\sqrt{\frac{2}{\pi t\epsilon}} \left(\int_{0}^{+\infty} e^{-\left(\frac{1}{2t}(x_i + y_i)^2 + \lambda_i y_i\right)/\epsilon} \, dy_i + \int_{0}^{+\infty} e^{-\left(\frac{1}{2t}(x_i - y_i)^2 + \lambda_i y_i\right)/\epsilon} \, dy_i \right) \right) \end{aligned}$$

To compute this integral, first define the function

$$\mathbb{R} \ni z \mapsto L(z) = \frac{1}{2} e^{z^2} \operatorname{erfc}(\mathrm{z}),$$

where erfc *denotes the complementary error function. Then we have ([20], page 336, integral 3.332, 2., and page 887, integral 8.250, 1.)*

$$\frac{1}{2}\sqrt{\frac{2}{\pi t\epsilon}} \int_{0}^{+\infty} e^{-\left(\frac{1}{2t}(x_i + y_i)^2 + \lambda_i y_i\right)/\epsilon} \, dy_i = e^{-\frac{x_i^2}{2t\epsilon}} L\left(\frac{x_i + t\lambda_i}{\sqrt{2t\epsilon}} \right)$$

and

$$\frac{1}{2}\sqrt{\frac{2}{\pi t\epsilon}} \int_{0}^{+\infty} e^{-\left(\frac{1}{2t}(x_i - y_i)^2 + \lambda_i y_i\right)/\epsilon} \, dy_i = e^{-\frac{x_i^2}{2t\epsilon}} L\left(\frac{-x_i + t\lambda_i}{\sqrt{2t\epsilon}} \right),$$

from which we get

$$S_\epsilon(\boldsymbol{x},t) = \frac{\|\boldsymbol{x}\|_2^2}{2t} - \epsilon \sum_{i=1}^{n} \ln \left(L\left(\frac{x_i + t\lambda_i}{\sqrt{2t\epsilon}} \right) + L\left(\frac{-x_i + t\lambda_i}{\sqrt{2t\epsilon}} \right) \right).$$

Now, to compute the posterior mean estimate it suffices to compute the gradient of $\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t)$ *and use the formula* $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) = \boldsymbol{x} - t\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t)$. *To do so, we must compute the derivative of the function* L. *Since*

$$\frac{dL}{dz}(z) = 2zL(z) + \frac{1}{\sqrt{\pi}},$$

the chain rule gives

$$\frac{\partial}{\partial x_i} \left(L\left(\frac{x_i + t\lambda_i}{\sqrt{2t\epsilon}} \right) + L\left(\frac{-x_i + t\lambda_i}{\sqrt{2t\epsilon}} \right) \right) = \left(\frac{x_i + t\lambda_i}{t\epsilon} \right) L\left(\frac{x_i + t\lambda_i}{\sqrt{2t\epsilon}} \right) - \left(\frac{-x_i + t\lambda_i}{t\epsilon} \right) L\left(\frac{-x_i + t\lambda_i}{\sqrt{2t\epsilon}} \right).$$

The posterior mean estimate is therefore given component-wise by

$$
\begin{aligned}
(\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon))_i &= x_i - t(\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t))_i \\
&= x_i + t\lambda_i \left(\frac{L\left(\frac{x_i+t\lambda_i}{\sqrt{2t\epsilon}}\right) + L\left(\frac{-x_i+t\lambda_i}{\sqrt{2t\epsilon}}\right)}{L\left(\frac{x_i+t\lambda_i}{\sqrt{2t\epsilon}}\right) - L\left(\frac{-x_i+t\lambda_i}{\sqrt{2t\epsilon}}\right)} \right)
\end{aligned}
$$

The posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)$ yields a smooth analogue of the soft thresholding operator T (defined in (38)) evaluated at $(x_i, t\lambda_i)$, in the sense that $\lim_{\substack{\epsilon\to 0\\ \epsilon>0}} (\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon))_i = T(x_i,t\lambda_i)$ for every $i \in \{1,\dots,n\}$ by Theorem 3.1(iv). Figure 2 shows the MAP and posterior mean estimates in one dimension for the choice of $t = 1.25$, $\epsilon = \{0.025,\, 0.1,\, 0.25,\, 0.5,\, 1\}$, and $\lambda_1 = 2$ for $x \in [-5,5]$.

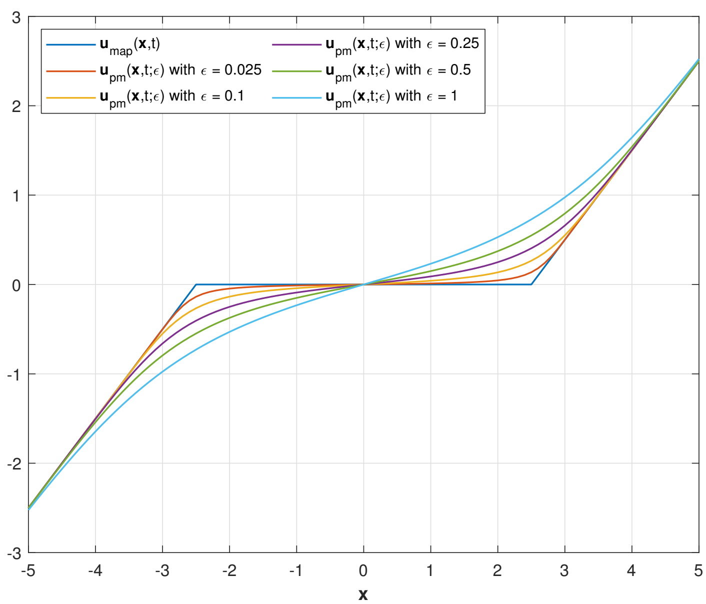

FIGURE 2. Numerical example of the MAP and posterior mean estimates in one dimension with the prior $J(x) = \lambda_1 |x|$ for the choice of $t = 1.25$, $\epsilon = \{0.025,\, 0.1,\, 0.25,\, 0.5,\, 1\}$, and $\lambda_1 = 2$ for $x \in [-5,5]$.

3.3. **Connections to first-order Hamilton–Jacobi equations.** In this section, we use the connections between the posterior mean estimate (5) and viscous HJ PDEs established in Theorem 3.1 to show that the posterior mean estimate can be expressed through the solution to a first-order HJ PDE with initial data of the form of (22). In particular, we show that the posterior mean estimate satisfies the proximal mapping formula

$$\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) = \operatorname*{arg\,min}_{\boldsymbol{y}\in\mathbb{R}^n} \left\{ \frac{1}{2}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + \left(K_\epsilon^*(\boldsymbol{y},t) - \frac{1}{2}\|\boldsymbol{y}\|_2^2 \right) \right\},$$

where the function $K_\epsilon \colon \mathbb{R}^n \times \mathbb{R}^n \times (0,+\infty)$ is defined through the solution $S_\epsilon(\boldsymbol{x},t)$ to the viscous HJ PDE (32) via

$$K_\epsilon(\boldsymbol{x},t) \coloneqq \frac{1}{2}\left\|\boldsymbol{x}\right\|_2^2 - tS_\epsilon(\boldsymbol{x},t) \equiv t\epsilon \ln\left(\frac{1}{(2\pi t\epsilon)^{n/2}} \int_{\text{dom } J} e^{\left(\frac{1}{t}\langle \boldsymbol{x},\boldsymbol{y}\rangle - \frac{1}{2t}\|\boldsymbol{y}\|_2^2 - J(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y} \right),$$

which is convex by Theorem 3.1(ii)(d), and where $K_\epsilon^*(\boldsymbol{y},t)$ denotes the Fenchel–Legendre transform of $\boldsymbol{y} \mapsto K_\epsilon(\boldsymbol{y},t)$.

This result gives the representation of the convex imaging prior whose existence was derived first, to our knowledge, by Louchet [34] and Gribonval [21] (and later extended to general Gaussian data fidelity terms in [22] and non-Gaussian data fidelity terms in [24, 23]).

[TODO: Mention Louchet's result that it is C2 - we show that our is C-infinity. Mention's Nikolova result - it is Cinfinity, so PM avoids staircasing based on her paper. Classic - now Cinf. Regularization results using EDPs. Update: I need to read Nikolova's paper in more detail, but I think only C2 is required - the minimizer is smoothing like C(m-1) if the prior is C(m)..]

Theorem 3.2 (Connections between the posterior mean estimate and first-order HJ PDEs)**.** *Suppose the function J satisfies assumptions (A1)-(A3). For every $\boldsymbol{x} \in \mathbb{R}^n$, $t > 0$, and $\epsilon > 0$, let $S_\epsilon(\boldsymbol{x},t)$ denote the solution to the second-order HJ PDE* (32) *with initial data J and let $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)$ denote the posterior mean estimate* (5)*. Consider the first-order HJ PDE*

$$(39)\qquad \begin{cases} \dfrac{\partial \tilde{S}}{\partial s}(\boldsymbol{x},s) + \dfrac{1}{2}\left\|\nabla_{\boldsymbol{x}}\tilde{S}(\boldsymbol{x},s)\right\|_2^2 = 0 & \text{in } \mathbb{R}^n \times (0,+\infty), \\ \tilde{S}(\boldsymbol{x},0) = K_\epsilon^*(\boldsymbol{x},t) - \dfrac{1}{2}\|\boldsymbol{x}\|_2^2 & \text{in } \mathbb{R}^n. \end{cases}$$

Then the initial data $\boldsymbol{x} \mapsto K_\epsilon^*(\boldsymbol{x},t) - \frac{1}{2}\|\boldsymbol{x}\|_2^2$ is convex, the solution to the HJ PDE (39) *satisfies the Lax–Oleinik formula*

$$\tilde{S}_0(\boldsymbol{x},s) = \inf_{\boldsymbol{y}\in\mathbb{R}^n} \left\{ \frac{1}{2s}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + \left(K_\epsilon^*(\boldsymbol{y},t) - \frac{1}{2}\|\boldsymbol{y}\|_2^2 \right) \right\},$$

and the corresponding minimizer at $s = 1$ is the posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)$:

$$\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) = \operatorname*{arg\,min}_{\boldsymbol{y}\in\mathbb{R}^n} \left\{ \frac{1}{2}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + \left(K_\epsilon^*(\boldsymbol{y},t) - \frac{1}{2}\|\boldsymbol{y}\|_2^2 \right) \right\}.$$

Proof. By definition of the function $(\boldsymbol{x}, t) \mapsto K_\epsilon(\boldsymbol{x}, t)$, we may write

$$tS_\epsilon(\boldsymbol{x}, t) + K_\epsilon(\boldsymbol{x}, t) = \frac{1}{2}\|\boldsymbol{x}\|_2^2.$$

As both $\boldsymbol{x} \mapsto tS_\epsilon(\boldsymbol{x}, t)$ and $\boldsymbol{x} \mapsto K_\epsilon(\boldsymbol{x}, t)$ are convex by Theorem 3.1(ii)(a) and (d), we can apply Moreau's decomposition theorem (see definition 7 in Section 2) to conclude that $\boldsymbol{x} \mapsto K_\epsilon^*(\boldsymbol{x}, t) - \frac{1}{2}\|\boldsymbol{x}\|_2^2$ is convex and to express $tS_\epsilon(\boldsymbol{x}, t)$ as

$$tS_\epsilon(\boldsymbol{x}, t) = \inf_{\boldsymbol{y} \in \mathbb{R}^n} \left\{ \frac{1}{2}\left\|\boldsymbol{x} - \boldsymbol{y}\right\|_2^2 + \left(K_\epsilon^*(\boldsymbol{y}, t) - \frac{1}{2}\left\|\boldsymbol{y}\right\|_2^2 \right) \right\} \tag{40}$$

On the one hand, by Theorem 2.1 the right hand side of (40) is the solution $\tilde{S}_0(\boldsymbol{x}, s)$ to the first-order HJ PDE (39) at $s = 1$, and therefore its minimizer is given by $\boldsymbol{x} - \nabla_{\boldsymbol{x}}\tilde{S}_0(\boldsymbol{x}, 1)$. On the other hand, the gradient $\nabla_{\boldsymbol{x}}\tilde{S}_0(\boldsymbol{x}, 1)$ is equal to the left hand side of (40), that is, $\nabla_{\boldsymbol{x}}\tilde{S}_0(\boldsymbol{x}, 1) = t\nabla_{\boldsymbol{x}}S_\epsilon(\boldsymbol{x}, t)$, which is equal to $\boldsymbol{x} - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$ by formula (5). As a result, the posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$ minimizes the right hand side of (40), that is,

$$\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) = \operatorname*{arg\,min}_{\boldsymbol{y} \in \mathbb{R}^n} \left\{ \frac{1}{2}\left\|\boldsymbol{x} - \boldsymbol{y}\right\|_2^2 + \left(K_\epsilon^*(\boldsymbol{y}, t) - \frac{1}{2}\|\boldsymbol{y}\|_2^2 \right) \right\}.$$

$\square$

4. Properties of MMSE and MAP estimators

In this section, we describe various properties of the Bayesian posterior mean estimate (5) in terms of the data $\boldsymbol{x} \in \mathbb{R}^n$, parameters $t > 0$ and $\epsilon > 0$, and the imaging prior J. Specifically, in Section 4.1, we derive topological, representation, and monotonicity properties of the posterior mean estimate, which we use in Section 4.2 to further derive various bounds and limit properties of the posterior mean estimate. Finally, we describe the Bregman divergence properties of both posterior mean and MAP estimates in Sect. 4.3.

4.1. **Topological, representation, and monotonicity properties.** This section describes the topological, representation, and monotonicity properties of the Bayesian posterior mean estimate (5), which are stated, respectively, in propositions 4.1, Theorem 4.2, and 4.3.

The first result, Proposition 4.1, states that the posterior mean estimate belongs in the interior of the domain of the prior J for all data $\boldsymbol{x} \in \mathbb{R}^n$ and parameters $t > 0$ and $\epsilon > 0$.

Proposition 4.1 (Topological properties)**.** *Suppose the function J satisfies assumptions (A1)-(A3), and let $\boldsymbol{x} \in \mathbb{R}^n$, $t > 0$, and $\epsilon > 0$. Then the posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$ is contained in* int dom J. *As a consequence, the subdifferential $\partial J(\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon))$ is non-empty and*

$$J(\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)) \leqslant \mathbb{E}_J\left[J(\boldsymbol{y})\right] < \epsilon\left(e^{S_\epsilon(\boldsymbol{x},t)/\epsilon} - 1\right) < +\infty.$$

Proof. See Appendix B. □

The second result, Proposition 4.1, gives representation formulas for the posterior mean estimate. In particular, when the prior satisfies assumptions (A1)-(A3) and dom $J = \mathbb{R}^n$, the posterior mean estimate and variance then satisfy representation formulas in terms of the mean minimal subgradient of J given by $\mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right]$. These representation formulas are then used to show that when dom $J \neq \mathbb{R}^n$, the posterior mean estimate can nonetheless be approximated using the first-order HJ PDE (22) by smoothing the initial J via its Moreau–Yosida approximation $S_0(\boldsymbol{x}, \mu)$ for any $\mu > 0$.

Proposition 4.2 (Representation properties)**.** *Suppose the function J satisfies assumptions (A1)-(A3), let $\boldsymbol{x} \in \mathbb{R}^n$, $t > 0$, and $\epsilon > 0$, and let $(\boldsymbol{x}, t) \mapsto S_0$ and $(\boldsymbol{x}, t) \mapsto S_\epsilon$ denote the solutions to the first and second-order HJ PDEs* (22) *and* (32) *with initial data J, respectively.*

(i) *(Representation formulas) Suppose that* dom $J = \mathbb{R}^n$. *Then the posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$ and the variance $\mathbb{E}_J\left[\|\boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)\|_2^2\right]$ of the Bayesian posterior distribution* (28) *satisfy the representation formulas*

$$\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) = \boldsymbol{x} - t\mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right] \tag{41}$$

and

$$\mathbb{E}_J\left[\|\boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)\|_2^2\right] = nt\epsilon - t\mathbb{E}_J\left[\left\langle \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0}), \boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)\right\rangle\right], \tag{42}$$

with $\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0}) = \nabla J(\boldsymbol{y})$ when J is continuously differentiable. In particular, the gradient and Laplacian of the solution $(\boldsymbol{x}, t) \mapsto S_\epsilon(\boldsymbol{x}, t)$ to the HJ PDE (32) *with initial data J satisfy*

$$\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x}, t) = \mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right]$$

and

$$\Delta_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x}, t) = \frac{1}{t\epsilon}\mathbb{E}_J\left[\left\langle \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0}), \boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)\right\rangle\right],$$

(ii) (Limit formulas) Let $\{\mu_k\}_{k=1}^{+\infty}$ *be a sequence of positive real numbers decreasing to zero. Then the gradient of the solution* $(\boldsymbol{x}, t) \mapsto S_\epsilon(\boldsymbol{x}, t)$ *to the HJ PDE* (32) *with initial data* J *satisfies the limit*

$$\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x}, t) = \lim_{k\to+\infty} \left(\frac{\int_{\mathbb{R}^n} \nabla_{\boldsymbol{y}} S_0(\boldsymbol{y}, \mu_k) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + S_0(\boldsymbol{y},\mu_k)\right)/\epsilon}\, d\boldsymbol{y}}{\int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + S_0(\boldsymbol{y},\mu_k)\right)/\epsilon}\, d\boldsymbol{y}} \right).$$

As a consequence, the posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$ *satisfies the limit*

$$\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) = \boldsymbol{x} - t \lim_{k\to+\infty} \left(\frac{\int_{\mathbb{R}^n} \nabla_{\boldsymbol{y}} S_0(\boldsymbol{y}, \mu_k) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + S_0(\boldsymbol{y},\mu_k)\right)/\epsilon}\, d\boldsymbol{y}}{\int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + S_0(\boldsymbol{y},\mu_k)\right)/\epsilon}\, d\boldsymbol{y}} \right). \tag{43}$$

Proof. See Appendix C for the proof. □

Remark 4.1. *Note that the representation formulas in Proposition 4.2(ii) may not hold if* dom $J \neq \mathbb{R}^n$. *To see this, consider the prior* $J : \mathbb{R}^n \mapsto \mathbb{R} \cup \{+\infty\}$ *defined by*

$$J(\boldsymbol{y}) = \begin{cases} 0, & \text{if } \|\boldsymbol{y}\|_2 \leqslant 1, \\ +\infty, & \text{otherwise.} \end{cases}$$

Then the domain of J *is the unit sphere in* $\mathbb{R}^n$, *which is convex, and* J *satisfies assumptions (A1)-(A3). The function* J *is continuously differentiable in* int dom J, *with* $\nabla J(\boldsymbol{y}) = \boldsymbol{0}$ *for every* $\boldsymbol{y} \in$ int dom J. *Clearly,* $\mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right] = 0$. *However, for every* $\boldsymbol{x} \neq \boldsymbol{0}$, *the posterior mean estimate* $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) \neq \boldsymbol{x}$. *Hence, the representation formula* (41) *does not hold in that case.*

The third result, Proposition 4.3, describes monotonicity properties of the posterior mean estimate, which in particular will be leveraged in the next subsection to derive an optimal upper bound on the variance $\mathbb{E}_J\left[\|\boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)\|_2^2\right]$ and several estimates and limit results of $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$ in terms of the observed image $\boldsymbol{x}$ and parameter $t > 0$. Our proof of the following proposition, which is presented in Appendix D, uses the properties of solutions to first-order HJ PDEs presented in Theorem 2.1 together with the representation formulas (41) and (42).

Proposition 4.3 (Monotonicity property)**.** *Suppose the function* J *satisfies assumptions (A1)-(A3), and let* $\boldsymbol{x} \in \mathbb{R}^n$, $t > 0$, *and* $\epsilon > 0$. *Suppose also that* J *is strongly convex of parameter* $m \geqslant 0$ *(with*

$m = 0$ corresponding to the definition of convexity). Then for every $\boldsymbol{y}_0 \in \operatorname{dom} \partial J$,

(44)
$$\left(\frac{1+mt}{t}\right)\mathbb{E}_J\left[\|\boldsymbol{y}-\boldsymbol{y}_0\|_2^2\right] \leqslant \mathbb{E}_J\left[\left\langle\left(\frac{\boldsymbol{y}-\boldsymbol{x}}{t}+\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right)-\left(\frac{\boldsymbol{y}_0-\boldsymbol{x}}{t}+\pi_{\partial J(\boldsymbol{y}_0)}(\boldsymbol{0})\right), \boldsymbol{y}-\boldsymbol{y}_0\right\rangle\right]$$
$$\leqslant n\epsilon - \left\langle\left(\frac{\boldsymbol{y}_0-\boldsymbol{x}}{t}+\pi_{\partial J(\boldsymbol{y}_0)}(\boldsymbol{0})\right), \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)-\boldsymbol{y}_0\right\rangle.$$

Proof. See Appendix D for the proof. □

As a corollary of this result, we now show that the mean minimal subgradient $\mathbb{E}_J\left[\mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right]\right]$ is finite; this result will be used later in Subsection 4.3 for proving Theorem 4.1(ii).

Corollary 4.1. *Suppose J satisfies assumptions (A1)-(A3). For every $\boldsymbol{x} \in \mathbb{R}^n$, $t > 0$, and $\epsilon > 0$, the mean minimal subgradient $\mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right]$ is finite.*

Proof. Let $\boldsymbol{y}_0$ be any element of int dom J different from $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)$; such an element exists because int dom $J \neq \varnothing$ by assumption (A2). Consider the scalar product

$$\left\langle\frac{\boldsymbol{y}-\boldsymbol{x}}{t}+\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0}), \boldsymbol{y}_0-\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)\right\rangle = \left\langle\left(\frac{\boldsymbol{y}-\boldsymbol{x}}{t}+\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right), \boldsymbol{y}-\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)\right\rangle$$
$$-\left\langle\left(\frac{\boldsymbol{y}-\boldsymbol{x}}{t}+\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right), \boldsymbol{y}-\boldsymbol{y}_0\right\rangle.$$

Taking the expected value $\mathbb{E}_J[\cdot]$ and using the monotonicity property (44), we get the inequality

$$-n\epsilon \leqslant \left\langle\left(\frac{\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)-\boldsymbol{x}}{t}+\mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right]\right)-\left(\frac{\boldsymbol{y}_0-\boldsymbol{x}}{t}+\pi_{\partial J(\boldsymbol{y}_0)}(\boldsymbol{0})\right), \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)-\boldsymbol{y}_0\right\rangle \leqslant n\epsilon.$$

The scalar product in the equation above is therefore finite, which implies that the mean minimal subgradient $\mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right]$ in the scalar product is also finite. □

4.2. **Bound and limit properties.** We now derive various bounds on the posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)$ and the variance $\mathbb{E}_J\left[\|\boldsymbol{y}-\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)\|_2^2\right]$ and limiting results of the posterior mean estimate in terms of the parameters t. Specifically, the next proposition presents an optimal upper bound for the variance, an estimate of the difference between the MAP and PM estimates, an estimate for the deviation between two posterior mean estimates $\boldsymbol{u}_{PM}(\boldsymbol{x}_1,t,\epsilon)$ and $\boldsymbol{u}_{PM}(\boldsymbol{x}_2,t,\epsilon)$, and the behavior of the posterior mean when the parameter t or ϵ goes to 0.

Proposition 4.4 (Bounds and limit properties)**.** *Suppose J satisfies assumptions (A1)-(A3), and suppose that it is strongly convex of parameter $m \geqslant 0$ (with $m = 0$ corresponding to the definition of convexity).*

(i) *For every* $\boldsymbol{x} \in \mathbb{R}^n$, $t > 0$, *and* $\epsilon > 0$, *the variance* $\mathbb{E}_J\left[\|\boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)\|_2^2\right]$ *of the Bayesian posterior distribution* (28) *satisfies the upper bound*

$$\mathbb{E}_J\left[\|\boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)\|_2^2\right] \leqslant \frac{nt\epsilon}{1 + mt}. \tag{45}$$

(ii) *For every* $\boldsymbol{x} \in \mathbb{R}^n$, $t > 0$, *and* $\epsilon > 0$, *the square of the Euclidean norm between the posterior mean estimate and the MAP estimate satisfies the upper bound*

$$\|\boldsymbol{u}_{MAP}(\boldsymbol{x}, t) - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)\|_2^2 \leqslant \frac{nt\epsilon}{1 + mt}. \tag{46}$$

(iii) *For every* $\boldsymbol{x}$, $\boldsymbol{d} \in \mathbb{R}^n$, $t > 0$, *and* $\epsilon > 0$,

$$\|\boldsymbol{u}_{PM}(\boldsymbol{x} + \boldsymbol{d}, t, \epsilon) - \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)\|_2 \leqslant \|\boldsymbol{d}\|_2. \tag{47}$$

(iv) *Let* $\{t_k\}_{k=1}^{+\infty}$ *be a sequence of positive real numbers converging to* 0 *and let* $\{\boldsymbol{d}_k\}_{k=1}^{+\infty}$ *be a sequence of elements of* $\mathbb{R}^n$ *converging to* $\boldsymbol{d} \in \mathbb{R}^n$. *Then for every* $\boldsymbol{x} \in \mathrm{dom}\ J$ *and* $\epsilon > 0$, *the pointwise limit of* $\boldsymbol{u}_{PM}(\boldsymbol{x} + t_k\boldsymbol{d}_k, t_k, \epsilon)$ *as* $k \to +\infty$ *exists and satisfies*

$$\lim_{k \to +\infty} \boldsymbol{u}_{PM}(\boldsymbol{x} + t_k\boldsymbol{d}_k, t_k, \epsilon) = \boldsymbol{x}.$$

Proof. Proof of (i): Since $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) \in \mathrm{int\ dom}\ J$ by Proposition 4.1 and $\mathrm{int\ dom}\ J \in \mathrm{dom}\ \partial J$ (see Definition 5), we can set $\boldsymbol{y}_0 = \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$ in the monotonicity inequality (44) in Proposition 4.3(i) and rearrange to get the upper bound (45).

Proof of (ii): Note that for every $\boldsymbol{y}_0 \in \mathrm{dom}\ \partial J$, the monotonicity inequality (44) in Proposition 4.3 yields

$$\mathbb{E}_J\left[\left\langle \left(\frac{\boldsymbol{y} - \boldsymbol{x}}{t} + \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right), \boldsymbol{y} - \boldsymbol{y}_0 \right\rangle\right] \leqslant n\epsilon.$$

Choose $\boldsymbol{y}_0 = \boldsymbol{u}_{MAP}(\boldsymbol{x}, t)$, which for every $\boldsymbol{x}$ and $t > 0$ is always an element of $\mathrm{dom}\ \partial J$ and also satisfies the inclusion $\left(\frac{\boldsymbol{x} - \boldsymbol{u}_{MAP}(\boldsymbol{x}, t)}{t}\right) \in \partial J(\boldsymbol{u}_{MAP}(\boldsymbol{x}, t))$ by part (ii) of Theorem 2.1. Hence the monotonicity of the subdifferential of $\boldsymbol{y} \mapsto \frac{1}{2t}\|\boldsymbol{x} - \boldsymbol{y}\|_2^2 + J(\boldsymbol{y})$ and strong convexity of J of parameter $m \geqslant 0$ implies

$$\left(\frac{1 + mt}{t}\right)\|\boldsymbol{y} - \boldsymbol{u}_{MAP}(\boldsymbol{x}, t)\|_2^2 \leqslant \left\langle \left(\frac{\boldsymbol{x} - \boldsymbol{y}}{t} + \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right), \boldsymbol{y} - \boldsymbol{u}_{MAP}(\boldsymbol{x}, t) \right\rangle.$$

Combine these inequalities to get $\mathbb{E}_J\left[\|\boldsymbol{y}-\boldsymbol{u}_{MAP}(\boldsymbol{x},t)\|_2^2\right] \leqslant \frac{nt\epsilon}{1+mt}$, and use the convexity of the Euclidean norm to get inequality (46).

Proof of (iii): Since the two functions $\boldsymbol{x} \mapsto S_\epsilon(\boldsymbol{x},t)$ and $\boldsymbol{x} \mapsto \frac{1}{2}\|\boldsymbol{x}\|_2^2 - tS_\epsilon(\boldsymbol{x},t)$ are convex by Theorem 3.1(ii)(a) and (d), the gradient of the function $\boldsymbol{x} \mapsto \frac{1}{2}\|\boldsymbol{x}\|_2^2 - tS_\epsilon(\boldsymbol{x},t)$, whose value is the posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)$ by Theorem 3.1(iii), is Lipschitz continuous with unit constant (see [54] for a simple proof). Hence, given $\boldsymbol{x}$, $\boldsymbol{d} \in \mathbb{R}^n$, $t > 0$

$$\|(\boldsymbol{x}+\boldsymbol{d}-t\nabla_{\boldsymbol{x}}S_\epsilon(\boldsymbol{x}+\boldsymbol{d},t)) - (\boldsymbol{x}-t\nabla_{\boldsymbol{x}}S_\epsilon(\boldsymbol{x},t))\|_2 \equiv \|\boldsymbol{u}_{PM}(\boldsymbol{x}+\boldsymbol{d},t,\epsilon) - \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)\|_2 \leqslant \|\boldsymbol{d}\|_2 .$$

Proof of (iv): Inequality (46) and the triangle inequality imply

$$\|(\boldsymbol{x}+t_k\boldsymbol{d}_k) - \boldsymbol{u}_{PM}(\boldsymbol{x}+t_k\boldsymbol{d}_k,t_k,\epsilon)\|_2 \leqslant \|(\boldsymbol{x}+t_k\boldsymbol{d}_k) - \boldsymbol{u}_{MAP}(\boldsymbol{x}+t_k\boldsymbol{d}_k,t_k)\|_2 + \sqrt{\frac{nt_k\epsilon}{1+mt}}.$$

The limit $\lim_{k\to+\infty}\boldsymbol{u}_{PM}(\boldsymbol{x}+t_k\boldsymbol{d}_k,t_k,\epsilon) = \boldsymbol{x}$ then follows by Theorem 2.1(i). □

Remark 4.2. *The upper bound for the variance in* (45) *is optimal; as shown in Example 1 it is attained for the quadratic prior* $J(\boldsymbol{x}) = \frac{m}{2}\|\boldsymbol{x}\|_2^2$.

4.3. **Bayesian decision theory, Bregman divergences, and Hamilton–Jacobi partial differential equations.** In this section, we will consider the Bayesian risks associated to the two Bregman loss functions

$$\boldsymbol{y} \mapsto D_{\Phi_J}(\boldsymbol{y},\varphi_J(\boldsymbol{u}|\boldsymbol{x},t)) \tag{48}$$

and

$$\boldsymbol{y} \mapsto D_{\Phi_J}(\boldsymbol{u},\varphi_J(\boldsymbol{y}|\boldsymbol{x},t)), \tag{49}$$

where

$$\mathbb{R}^n \times \mathbb{R}^n \times (0,+\infty) \ni (\boldsymbol{y},\boldsymbol{x},t) \mapsto \Phi_J(\boldsymbol{y}|\boldsymbol{x},t) = \frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y}),$$

which is up to a constant the negative logarithm of the posterior distribution (28), and

$$\mathbb{R}^n \times \mathbb{R}^n \times (0,+\infty) \ni (\boldsymbol{y},\boldsymbol{x},t) \mapsto \varphi_J(\boldsymbol{y}|\boldsymbol{x},t) = \left(\frac{\boldsymbol{y}-\boldsymbol{x}}{t}\right) + \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0}),$$

which is a subgradient of the function $\boldsymbol{y} \mapsto \Phi_J(\boldsymbol{y}|\boldsymbol{x},t)$. The corresponding Bayesian risks to the posterior distribution (28) correspond to the expected values $\mathbb{E}_J\left[D_{\Phi_J}(\boldsymbol{y},\varphi_J(\boldsymbol{u}|\boldsymbol{x},t))\right]$ and $\mathbb{E}_J\left[D_{\Phi_J}(\boldsymbol{u},\varphi_J(\boldsymbol{y}|\boldsymbol{x},t))\right]$, respectively. We refer the reader to [3] and [29] for discussions on Bregman loss functions and Bayesian estimation theory.

Recent work by Burger and Lucka [5] has shown that the MAP estimate (2) corresponds to the Bayes estimator associated to the Bregman loss function (49) when the prior is convex and uniformly Lipschitz continuous. This was later extended by M. Burger and Sciacchitano [36] to posterior distributions with non-Gaussian fidelity term, and later studied from the point of view from differential geometry by Pereyra [40] and also derived for strongly log-concave and posterior distributions. Here, we will improve these results and show that they follow naturally from the connections between maximum a posteriori and posterior mean estimates and Hamilton–Jacobi equations derived in Section 3. In particular, we show that when the prior J is convex and finite on $\mathbb{R}^n$, then the MAP estimate $\boldsymbol{u}_{MAP}(\boldsymbol{x},t)$ minimizes in expectation the Bregman loss function (49). This generalizes the result from Burger and Lucka [5] (Theorem 1), at least when the fidelity term takes the form $\frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2$, in that the prior needs not be uniformly Lipschitz continuous, and the proof that we present below is based on the connections between the MAP and posterior mean estimates and HJ PDEs derived earlier in Section 3. We also show that when dom $J \neq \mathbb{R}^n$, there still exists a Bayes estimator. A similar result was established in Pereyra [40] (see Theorem 4 and section 5.3), where the prior was assumed to be thrice differentiable and under strong convexity assumptions on the posterior distribution. In contrast, our results only need that J satisfies assumptions (A1)-(A3). The results rely on the monotonicity property (44) and finiteness of the mean minimal subgradient $\mathbb{E}_J\left[\pi_{\partial}(\boldsymbol{y})(\boldsymbol{0})\right]$ as shown in Corollary 4.1.

Theorem 4.1 (Bregman divergences)**.** *Suppose the function J satisfies assumptions (A1)-(A3), and let $\boldsymbol{x} \in \mathbb{R}^n$, $t > 0$, and $\epsilon > 0$.*

(i) The posterior mean estimate $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)$ is the unique global minimizer of the Bregman loss function dom $\partial J \ni \boldsymbol{u} \mapsto \mathbb{E}_J\left[D_{\Phi_J}(\boldsymbol{y},\varphi_J(\boldsymbol{u}|\boldsymbol{x},t))\right] \in \mathbb{R}$*, i.e.,*

$$\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) = \underset{\boldsymbol{u}\in \mathrm{dom}\ \partial J}{\arg\min}\ \mathbb{E}_J\left[D_{\Phi_J}(\boldsymbol{y},\varphi_J(\boldsymbol{u},\boldsymbol{x},t))\right]. \tag{50}$$

(ii) The mean Bregman loss function dom $J \ni \boldsymbol{u} \mapsto \mathbb{E}_J\left[D_{\Phi_J}(\boldsymbol{u},\varphi_J(\boldsymbol{y}|\boldsymbol{x},t))\right] \in \mathbb{R}$ *has a unique minimizer $\bar{\boldsymbol{u}} \in$ dom ∂J that satisfies the inclusion*

$$\left(\frac{\boldsymbol{x}-\bar{\boldsymbol{u}}}{t}\right) \in \partial J(\bar{\boldsymbol{u}}) + \left(\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t) - \mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right]\right), \tag{51}$$

where addition in (51) *is taken in the sense of sets.*

(iii) If J is finite everywhere on $\mathbb{R}^n$, i.e., $\mathrm{dom}\ J = \mathbb{R}^n$*, then the MAP estimate $\boldsymbol{u}_{MAP}(\boldsymbol{x}, t)$ is the unique global minimizer of the Bregman loss function $\mathbb{R}^n \ni \boldsymbol{u} \mapsto \mathbb{E}_J\left[D_{\Phi_J}(\boldsymbol{u}, \varphi_J(\boldsymbol{y}|\boldsymbol{x}, t))\right] \in \mathbb{R}$, i.e.,*

$$\boldsymbol{u}_{MAP}(\boldsymbol{x}, t) = \underset{\boldsymbol{u}\in\mathbb{R}^n}{\arg\min}\, \mathbb{E}_J\left[D_{\Phi_J}(\boldsymbol{u}, \varphi_J(\boldsymbol{y}, \boldsymbol{x}, t))\right] \tag{52}$$

Proof. See Appendix E for the proof.

□

5. Conclusion

[TODO: Summary of the contributions + Future perspectives. Discuss 1) extensions to the general class of convex variational problems discussed in [10] and 2) Exact formula for sampling via the minimal]

[TODO: In the discussion section, say that we anticipate the results to hold for general posterior distributions (as also shown by [36] under stronger assumptions), and we expect to establish these results in a forthcoming paper. Also, these results will be established using HJ equations.]

Appendix A. Proof of Theorem 3.1

To prove Theorem 3.1, we will first use the following lemma, which characterizes the partition function (29) in terms of the solution to a Cauchy problem involving the heat equation with initial data $J \in \Gamma_0(\mathbb{R}^n)$. This connection will imply parts (i) and (ii)(a)-(d) of Theorem 3.1.

Lemma A.1 (The heat equation with initial data in $\Gamma_0(\mathbb{R}^n)$)**.** *Suppose the function J satisfies assumptions (A1)-(A3).*

(i) For every $\epsilon > 0$, the function $w_\epsilon \colon \mathbb{R}^n \times [0, +\infty) \mapsto (0, 1]$ defined by

$$w_\epsilon(\boldsymbol{x}, t) := \frac{1}{(2\pi t\epsilon)^{n/2}} Z_J(\boldsymbol{x}, t, \epsilon) = \frac{1}{(2\pi t\epsilon)^{n/2}} \int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon} \, d\boldsymbol{y} \tag{53}$$

is the unique smooth solution to the Cauchy problem

$$\begin{cases} \dfrac{\partial w_\epsilon}{\partial t}(\boldsymbol{x}, t) = \dfrac{\epsilon}{2} \Delta_{\boldsymbol{x}} w_\epsilon(\boldsymbol{x}, t) & \textit{in } \mathbb{R}^n \times (0, +\infty), \\ w_\epsilon(\boldsymbol{x}, 0) = e^{-J(\boldsymbol{x})/\epsilon} & \textit{in } \mathbb{R}^n. \end{cases} \tag{54}$$

In addition, the domain of integration of the integral (53) *can be taken to be* dom J *or, up to a set of Lebesgue measure zero,* int (dom J) *or* dom ∂J*. Furthermore, for every $\boldsymbol{x} \in$ dom J and $\epsilon > 0$, except possibly at the boundary points $\boldsymbol{x} \in (\text{dom } J) \setminus (\text{int } (\text{dom } J))$ if such points exist, the pointwise limit of $w_\epsilon(\boldsymbol{x}, t)$ as $t \to 0$ exists and satisfies*

$$\lim_{\substack{t \to 0 \\ t > 0}} w_\epsilon(\boldsymbol{x}, t) = e^{-J(\boldsymbol{x})/\epsilon},$$

with the limit equal to 0 *whenever $\boldsymbol{x} \notin$ dom J. If $\boldsymbol{x} \in (\text{dom } J) \setminus (\text{int } (\text{dom } J))$, then we may only conclude*

$$\limsup_{\substack{t \to 0 \\ t > 0}} w_\epsilon(\boldsymbol{x}, t) \leqslant e^{-J(\boldsymbol{x})/\epsilon}$$

and

$$\liminf_{\substack{t \to 0 \\ t > 0}} w_\epsilon(\boldsymbol{x}, t) \geqslant e^{-J(\boldsymbol{x})/\epsilon} \left(\frac{1}{(2\pi\epsilon)^{n/2}} \int_{\text{dom } J} e^{-\frac{1}{2\epsilon}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2} \, d\boldsymbol{y} \right).$$

(ii) (Log-concavity and monotonicity properties).

(a) The function $\mathbb{R}^n \times (0, +\infty) \ni (\boldsymbol{x}, t) \mapsto t^{n/2} w_\epsilon(\boldsymbol{x}, t)$ is jointly strictly log-concave.

(b) The function $(0, +\infty) \ni t \mapsto t^{n/2} w_\epsilon(\boldsymbol{x}, t)$ is strictly monotone increasing.

(c) The function $(0, +\infty) \ni \epsilon \mapsto \epsilon^{n/2} w_\epsilon(\boldsymbol{x}, t)$ is strictly monotone increasing.

(d) The function $\mathbb{R}^n \ni \boldsymbol{x} \mapsto e^{\frac{1}{2t\epsilon}\|\boldsymbol{x}\|_2^2} w_\epsilon(\boldsymbol{x}, t)$ is log-convex.

The proof of (i) follows from classical PDEs arguments for the Cauchy problem (54) tailored to the initial data $(\boldsymbol{x}, \epsilon) \mapsto e^{-J(\boldsymbol{x})/\epsilon}$ with J satisfying assumptions (A1)-(A3), and the proof of log-concavity and monotonicity (ii)(a)-(d) follows from the Prékopa–Leindler and Hölder's inequalities [32, 43, 18]; we present the details below.

Proof. Proof of Lemma A.1 (i): By assumptions (A1) and (A2), there exists a point $\boldsymbol{y}_0 \in \text{int}\ (\text{dom}\ J)$ and a number $\delta > 0$ such that the open ball $B_\delta(\boldsymbol{y}_0)$ is contained in int (dom J) and $e^{-J(\boldsymbol{y})/\epsilon} > 0$ whenever $\boldsymbol{y} \in B_\delta(\boldsymbol{y}_0)$. Since assumption (A3) yields $e^{-J(\boldsymbol{y})/\epsilon} \leqslant 1$ for every $\boldsymbol{y} \in \mathbb{R}^n$, these observations imply that the Cauchy problem (54) has a unique, smooth solution defined by equation (53), with $0 < w_\epsilon(\boldsymbol{x}, t) \leqslant 1$ for every $\boldsymbol{x} \in \mathbb{R}^n$, $t > 0$, and $\epsilon > 0$ (see Widder [51], Theorem 1 in Chapter VII for global existence and smoothness properties of solutions, and Theorem 2.2 in Chapter VIII for uniqueness of solutions, and note that the results in [51] are for $n = 1$ but can be extended without difficulty to $n > 1$). In addition, as $e^{-J(\boldsymbol{y})/\epsilon} = 0$ for every $\boldsymbol{y} \notin \text{dom}\ J$, the domain of integration of (53) can be taken to be dom J, as the boundary points of the domain of J is a set of Lebesgue measure zero relative to $\mathbb{R}^n$ (see definition 2), the domain of integration can be further taken to be int (dom J) or dom ∂J.

Now, we will use Fatou's lemma ([18], Lemma 2.18) to compute bounds for the two limits $\limsup_{\substack{t\to 0\\ t>0}} w_\epsilon(\boldsymbol{x}, t)$ and $\liminf_{\substack{t\to 0\\ t>0}} w_\epsilon(\boldsymbol{x}, t)$ for every $\boldsymbol{x} \in \mathbb{R}^n$ and $\epsilon > 0$. First, note that the change of variables $\frac{\boldsymbol{x}-\boldsymbol{y}}{\sqrt{t}} \mapsto \boldsymbol{y}$ in (53) yields

$$w_\epsilon(\boldsymbol{x}, t) = \frac{1}{(2\pi\epsilon)^{n/2}} \int_{\mathbb{R}^n} e^{-\left(\frac{1}{2}\|\boldsymbol{y}\|_2^2 + J(\boldsymbol{x} - \sqrt{t}\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y},$$

so that the function

$$\boldsymbol{y} \mapsto e^{-\frac{1}{2\epsilon}\|y\|_2^2}\left(1 - e^{-J(\boldsymbol{x}-\sqrt{t}\boldsymbol{y})/\epsilon}\right)$$

is non-negative. Fatou's lemma therefore also applies to this function, and hence

$$\limsup_{\substack{t\to 0\\ t>0}} w_\epsilon(\boldsymbol{x}, t) \leqslant \frac{1}{(2\pi\epsilon)^{n/2}} \int_{\mathbb{R}^n} e^{-\frac{1}{2\epsilon}\|\boldsymbol{y}\|_2^2} \left(\limsup_{\substack{t\to 0\\ t>0}} e^{-J(\boldsymbol{x}-\sqrt{t}\boldsymbol{y})/\epsilon}\right) d\boldsymbol{y}.$$

Using the lower semicontinuity of J, the limit inside the integral satisfies

$$\begin{aligned}\limsup_{\substack{t\to 0\\ t>0}} e^{-J(\boldsymbol{x}-\sqrt{t}\boldsymbol{y})/\epsilon} &= e^{\limsup_{\substack{t\to 0\\ t>0}} -J(\boldsymbol{x}-\sqrt{t}\boldsymbol{y})/\epsilon} \\ &= e^{-\liminf_{\substack{t\to 0\\ t>0}} J(\boldsymbol{x}-\sqrt{t}\boldsymbol{y})/\epsilon} \\ &\leqslant e^{-J(\boldsymbol{x})/\epsilon}, \quad \forall \boldsymbol{y}\in\mathbb{R}^n,\end{aligned}$$

and therefore $\limsup_{\substack{t\to 0\\ t>0}} w_\epsilon(\boldsymbol{x},t) \leqslant e^{-J(\boldsymbol{x})/\epsilon}$ for every $\boldsymbol{x}\in\mathbb{R}^n$. If $\boldsymbol{x}\notin \operatorname{dom} J$, then

$$0 \leqslant \liminf_{\substack{t\to 0\\ t>0}} e^{-J(\boldsymbol{x}-\sqrt{t}\boldsymbol{y})/\epsilon} \leqslant \limsup_{\substack{t\to 0\\ t>0}} e^{-J(\boldsymbol{x}-\sqrt{t}\boldsymbol{y})/\epsilon} \leqslant 0,$$

which implies $\lim_{\substack{t\to 0\\ t>0}} w_\epsilon(\boldsymbol{x},t) = 0$ for every $\boldsymbol{x}\notin \operatorname{dom} J$. Suppose now that $\boldsymbol{x}\in\operatorname{dom} J$. By Fatou's lemma,

$$\liminf_{\substack{t\to 0\\ t>0}} w_\epsilon(\boldsymbol{x},t) \geqslant \frac{1}{(2\pi\epsilon)^{n/2}}\int_{\mathbb{R}^n} e^{-\frac{1}{2\epsilon}\|\boldsymbol{y}\|_2^2}\left(\liminf_{\substack{t\to 0\\ t>0}} e^{-J(\boldsymbol{x}-\sqrt{t}\boldsymbol{y})/\epsilon}\right) d\boldsymbol{y}.$$

If $\boldsymbol{x}\in\operatorname{int}(\operatorname{dom} J)$, then by continuity $\lim_{\substack{t\to 0\\ t>0}} J(\boldsymbol{x}-\sqrt{t}\boldsymbol{y}) = J(\boldsymbol{x})$ for every $\boldsymbol{y}\in\mathbb{R}^n$, and so $\liminf_{\substack{t\to 0\\ t>0}} w_\epsilon(\boldsymbol{x},t) \geqslant e^{-J(\boldsymbol{x})/\epsilon}$. Combined with the lim sup, we find $\lim_{\substack{t\to 0\\ t>0}} w_\epsilon(\boldsymbol{x},t) = e^{-J(\boldsymbol{x})/\epsilon}$ for every $\boldsymbol{x}\in\operatorname{int}(\operatorname{dom} J)$ and $\boldsymbol{x}\notin\operatorname{dom} J$. If $\boldsymbol{x}\in(\operatorname{dom} J)\backslash\operatorname{int}(\operatorname{dom} J)$, if any such point exists, and $0<t\leqslant 1$, then convexity of J implies

$$\begin{aligned}J(\boldsymbol{x}-\sqrt{t}\boldsymbol{y}) &= J((1-\sqrt{t})\boldsymbol{x}+\sqrt{t}(\boldsymbol{x}-\boldsymbol{y})) \\ &\leqslant (1-\sqrt{t})J(\boldsymbol{x}) + \sqrt{t}J(\boldsymbol{x}-\boldsymbol{y}),\end{aligned}$$

and hence for any $0<t\leqslant 1$,

$$\begin{aligned}w_\epsilon(\boldsymbol{x},t) &\geqslant e^{-(1-\sqrt{t})J(\boldsymbol{x})/\epsilon}\frac{1}{(2\pi\epsilon)^{n/2}}\int_{\mathbb{R}^n} e^{-\frac{1}{2\epsilon}\|\boldsymbol{y}\|_2^2} e^{-\sqrt{t}J(\boldsymbol{x}-\boldsymbol{y})/\epsilon}\, d\boldsymbol{y}, \\ &= e^{-(1-\sqrt{t})J(\boldsymbol{x})/\epsilon}\frac{1}{(2\pi\epsilon)^{n/2}}\int_{\mathbb{R}^n} e^{-\frac{1}{2\epsilon}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2} e^{-\sqrt{t}J(\boldsymbol{y})/\epsilon}\, d\boldsymbol{y}.\end{aligned}$$

Since $\liminf_{\substack{t\to 0\\ t>0}} e^{-\sqrt{t}J(\boldsymbol{y})/\epsilon} = 1$ for every $\boldsymbol{y}\in\operatorname{dom} J$ and is equal to 0 for every $\boldsymbol{y}\notin\operatorname{dom} J$, Fatou's lemma yields

$$\liminf_{\substack{t\to 0\\ t>0}} w_\epsilon(\boldsymbol{x},t) \geqslant e^{-J(\boldsymbol{x})/\epsilon}\left(\frac{1}{(2\pi\epsilon)^{n/2}}\int_{\operatorname{dom}\, J} e^{-\frac{1}{2\epsilon}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2}\, d\boldsymbol{y}\right)$$

for every $\boldsymbol{x}\in(\operatorname{dom} J)\backslash\operatorname{int}(\operatorname{dom} J)$, if any such point exists.

Proof of Lemma A.1 (ii)(a): The log-concavity property will be shown using the Prékopa–Leindler inequality.

Theorem A.1. *[Prékopa–Leindler inequality [32, 43]] Let f, g, and h be non-negative real-valued and Borel measurable functions on $\mathbb{R}^n$, and suppose*

$$h(\lambda \boldsymbol{y}_1 + (1-\lambda)\boldsymbol{y}_2) \geqslant f(\boldsymbol{y}_1)^{\lambda} g(\boldsymbol{y}_2)^{(1-\lambda)}$$

for every $\boldsymbol{y}_1$, $\boldsymbol{y}_2 \in \mathbb{R}^n$ and $\lambda \in (0,1)$. Then

$$\int_{\mathbb{R}^n} h(\boldsymbol{y})\, d\boldsymbol{y} \geqslant \left(\int_{\mathbb{R}^n} f(\boldsymbol{y})\, d\boldsymbol{y}\right)^{\lambda} \left(\int_{\mathbb{R}^n} g(\boldsymbol{y})\, d\boldsymbol{y}\right)^{(1-\lambda)}.$$

Let $\epsilon > 0$, $\lambda \in [0,1]$, $\boldsymbol{x} = \lambda \boldsymbol{x}_1 + (1-\lambda)\boldsymbol{x}_2$, $\boldsymbol{y} = \lambda \boldsymbol{y}_1 + (1-\lambda)\boldsymbol{y}_2$, and $t = \lambda t_1 + (1-\lambda)t_2$ for any $\boldsymbol{x}_1$, $\boldsymbol{x}_2$, $\boldsymbol{y}_1$, $\boldsymbol{y}_2 \in \mathbb{R}^n$ and t_1, $t_2 \in (0,+\infty)$. The joint convexity of the function $\mathbb{R}^n \times (0,+\infty) \ni (\boldsymbol{z}, t) \mapsto \frac{1}{2t}\left\|\boldsymbol{z}\right\|_2^2$ and convexity of J imply

$$\frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + J(\boldsymbol{y}) \leqslant \frac{\lambda}{2t_1}\left\|\boldsymbol{x}_1-\boldsymbol{y}_1\right\|_2^2 + \frac{(1-\lambda)}{2t_2}\left\|\boldsymbol{x}_2-\boldsymbol{y}_2\right\|_2^2 + \lambda J(\boldsymbol{y}_1) + (1-\lambda)J(\boldsymbol{y}_2),$$

This gives

$$\frac{e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}}{(2\pi\epsilon)^{n/2}} \geqslant \left(\frac{e^{-\left(\frac{1}{2t_1}\|\boldsymbol{x}_1-\boldsymbol{y}_1\|_2^2 + J(\boldsymbol{y}_1)\right)/\epsilon}}{(2\pi\epsilon)^{n/2}}\right)^{\lambda} \left(\frac{e^{-\left(\frac{1}{2t_2}\|\boldsymbol{x}_2-\boldsymbol{y}_2\|_2^2 + J(\boldsymbol{y}_2)\right)/\epsilon}}{(2\pi\epsilon)^{n/2}}\right)^{1-\lambda}.$$

Applying the Prékopa–Leindler inequality with

$$h(\boldsymbol{y}) = \frac{e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}}{(2\pi\epsilon)^{n/2}},$$

$$f(\boldsymbol{y}) = \frac{e^{-\left(\frac{1}{2t_1}\|\boldsymbol{x}_1-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}}{(2\pi\epsilon)^{n/2}},$$

and

$$g(\boldsymbol{y}) = \frac{e^{-\left(\frac{1}{2t_2}\|\boldsymbol{x}_2-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}}{(2\pi\epsilon)^{n/2}},$$

and using the definition (53) of $w_\epsilon(\boldsymbol{x}, t)$, we get

$$t^{n/2} w_\epsilon(\boldsymbol{x}, t) \geqslant \left(t_1^{n/2} w_\epsilon(\boldsymbol{x}_1, t_1)\right)^{\lambda} \left(t_2^{n/2} w_\epsilon(\boldsymbol{x}_2, t_2)\right)^{(1-\lambda)},$$

As a result, the function $(\boldsymbol{x}, t) \mapsto t^{n/2} w_\epsilon(\boldsymbol{x}, t)$ is jointly log-concave on $\mathbb{R}^n \times (0,+\infty)$.

Proof of Lemma A.1 (ii)(b): Since $t \mapsto \frac{1}{t}$ is strictly monotone decreasing on $(0, +\infty)$, for any $\boldsymbol{x} \in \mathbb{R}^n$, $\epsilon > 0$, and $0 < t_1 < t_2$,

$$\frac{e^{-\left(\frac{1}{2t_1}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon}}{(2\pi\epsilon)^{n/2}} < \frac{e^{-\left(\frac{1}{2t_2}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon}}{(2\pi\epsilon)^{n/2}}$$

whenever $\boldsymbol{x} \neq \boldsymbol{y}$. Integrating both sides of the inequality with respect to $\boldsymbol{y}$ over dom J yields

$$\frac{1}{(2\pi\epsilon)^{n/2}}\int_{\text{dom } J} e^{-\left(\frac{1}{2t_1}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y} < \frac{1}{(2\pi\epsilon)^{n/2}}\int_{\text{dom } J} e^{-\left(\frac{1}{2t_2}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y},$$

As a result, the function $t \mapsto t^{n/2} w_\epsilon(\boldsymbol{x}, t)$ is strictly monotone increasing on $(0, +\infty)$.

Proof of Lemma A.1 (ii)(c): Since $\epsilon \mapsto \frac{1}{\epsilon}$ is strictly monotone decreasing on $(0, +\infty)$ and $\boldsymbol{y} \mapsto J(\boldsymbol{y})$ is non-negative by assumption (A3), for any $\boldsymbol{x} \in \mathbb{R}^n$, $t > 0$, and $0 < \epsilon_1 < \epsilon_2$,

$$e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon_1} < e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon_2}$$

whenever $\boldsymbol{x} \neq \boldsymbol{y}$. Integrating both sides of the inequality with respect to $\boldsymbol{y}$ over dom J yields

$$\int_{\text{dom } J} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon_1} d\boldsymbol{y} < \int_{\text{dom } J} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon_2}\, d\boldsymbol{y},$$

As a result, the function $\epsilon \mapsto \epsilon^{n/2} w_\epsilon(\boldsymbol{x}, t)$ is strictly monotone increasing on $(0, +\infty)$.

Proof of Lemma A.1 (ii)(d): Let $\epsilon > 0$, $t > 0$, $\lambda \in [0, 1]$, and $\boldsymbol{x} = \lambda\boldsymbol{x}_1 + (1-\lambda)\boldsymbol{x}_2$. Then

$$\begin{aligned} e^{\frac{1}{2t\epsilon}\|\boldsymbol{x}\|_2^2} w_\epsilon(\boldsymbol{x}, t) &= \frac{1}{(2\pi t\epsilon)^{n/2}}\int_{\text{dom } J} e^{(\langle \boldsymbol{x},\boldsymbol{y}\rangle/t - J(\boldsymbol{y}))/\epsilon}\, d\boldsymbol{y} \\ &= \int_{\text{dom } J} \left(\frac{e^{(\langle \boldsymbol{x}_1,\boldsymbol{y}\rangle/t - J(\boldsymbol{y}))/\epsilon}}{(2\pi t\epsilon)^{n/2}}\right)^{\lambda} \left(\frac{e^{\langle \boldsymbol{x}_2,\boldsymbol{y}\rangle/t\epsilon - J(\boldsymbol{y})/\epsilon}}{(2\pi t\epsilon)^{n/2}}\right)^{1-\lambda} d\boldsymbol{y}. \end{aligned}$$

Hölder's inequality ([18], theorem 6.2) then implies

$$\begin{aligned} e^{\frac{1}{2t\epsilon}\|\boldsymbol{x}\|_2^2} w_\epsilon(\boldsymbol{x}, t) &\leqslant \left(\int_{\text{dom } J} \frac{e^{(\langle \boldsymbol{x}_1,\boldsymbol{y}\rangle/t - J(\boldsymbol{y}))/\epsilon}}{(2\pi t\epsilon)^{n/2}}\, d\boldsymbol{y}\right)^{\lambda} \left(\int_{\text{dom } J} \frac{e^{\langle \boldsymbol{x}_2,\boldsymbol{y}\rangle/t\epsilon - J(\boldsymbol{y})/\epsilon}}{(2\pi t\epsilon)^{n/2}}\, d\boldsymbol{y}\right)^{1-\lambda} \\ &= \left(e^{\frac{1}{2t\epsilon}\|\boldsymbol{x}_1\|_2^2} w_\epsilon(\boldsymbol{x}_1, t)\right)^{\lambda} \left(e^{\frac{1}{2t\epsilon}\|\boldsymbol{x}_2\|_2^2} w_\epsilon(\boldsymbol{x}_2, t)\right)^{1-\lambda}. \end{aligned}$$

As a result, the function $\mathbb{R}^n \ni \boldsymbol{x} \mapsto e^{\frac{1}{2t\epsilon}\|\boldsymbol{x}\|_2^2} w_\epsilon(\boldsymbol{x}, t)$ is log-convex. □

Proof of Theorem 3.1 (i) and (ii)(a)-(d): The proof of these follow from Lemma A.1 and classic results about the Cole–Hopf transform (see, e.g., [16], Section 4.4.1), with $S_\epsilon(\boldsymbol{x}, t) \coloneqq -\epsilon \log(w_\epsilon(\boldsymbol{x}, t))$.

Proof of Theorem 3.1 (iii): The result follows from a straightforward calculation of the gradient, divergence, and Laplacian of $S_\epsilon(\boldsymbol{x},t)$ that we omit here.

Proof of Theorem 3.1 (iv): The proof we present here is based on techniques from large deviation theory in probability theory [13, 14, 49] tailored to equation (31) and is adapted from Lemmas 2.1.7 and 2.1.8 of Deuschel and Stroock's book on Large Deviations [14]. We proceed in three steps:

Step 1. Show that

$$\limsup_{\substack{\epsilon\to 0\\ \epsilon>0}} S_\epsilon(\boldsymbol{x},t) \leqslant \inf_{\boldsymbol{y}\in \mathrm{int}\ (\mathrm{dom}\ J)} \left\{\frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + J(\boldsymbol{y})\right\}$$

and

$$\inf_{\boldsymbol{y}\in \mathrm{int}\ (\mathrm{dom}\ J)} \left\{\frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + J(\boldsymbol{y})\right\} = \inf_{\boldsymbol{y}\in \mathrm{dom}\ J} \left\{\frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + J(\boldsymbol{y})\right\} = S_0(\boldsymbol{x},t).$$

Step 2. Show that $\liminf_{\substack{\epsilon\to 0\\ \epsilon>0}} S_\epsilon(\boldsymbol{x},t) \geqslant S_0(\boldsymbol{x},t)$.

Step 3. Conclude that $\lim_{\substack{\epsilon\to 0\\ \epsilon>0}} S_\epsilon(\boldsymbol{x},t) = S_0(\boldsymbol{x},t)$. Pointwise and local uniform convergence of the gradient $\lim_{\substack{\epsilon\to 0\\ \epsilon>0}} \nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t) = \nabla_{\boldsymbol{x}} S_0(\boldsymbol{x},t)$, the partial derivative $\lim_{\substack{\epsilon\to 0\\ \epsilon>0}} \frac{\partial S_\epsilon(\boldsymbol{x},t)}{\partial t} = \frac{\partial S_0(\boldsymbol{x},t)}{\partial t}$, and the Laplacian $\lim_{\substack{\epsilon\to 0\\ \epsilon>0}} \frac{\epsilon}{2}\Delta_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t) = 0$ then follow from the convexity and differentiability of the solutions $(\boldsymbol{x},t)\mapsto S_0(\boldsymbol{x},t)$ and $(\boldsymbol{x},t)\mapsto S_\epsilon(\boldsymbol{x},t)$ to the HJ PDEs (22) and (32).

We will use the following large deviation principle [14]: For every Lebesgue measurable set $\mathcal{A}\in\mathbb{R}^n$,

$$\lim_{\substack{\epsilon\to 0\\ \epsilon>0}} -\epsilon\ln\left(\frac{1}{(2\pi t\epsilon)^{n/2}}\int_{\mathcal{A}} e^{-\frac{1}{2t\epsilon}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2}\, d\boldsymbol{y}\right) = \operatorname*{ess\,inf}_{\boldsymbol{y}\in\mathcal{A}} \frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2,$$

where essential infimum means infimum that holds almost everywhere, that is, if the essential infimum above is attained at $\boldsymbol{a}\in\mathcal{A}$, then $\left\|\boldsymbol{x}-\boldsymbol{a}\right\|_2^2/2t \leqslant \left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2/2t$ for almost every $\boldsymbol{y}\in\mathcal{A}$.

Step 1. (Adapted from Deuschel and Stroock [14], Lemma 2.1.7.) By convexity, the function J is continuous for every $\boldsymbol{y}_0\in \mathrm{int}\ (\mathrm{dom}\ J)$, the latter set being open. Therefore, for every such $\boldsymbol{y}_0$ there exists a number $r_{\boldsymbol{y}_0}>0$ such that for every $0<r\leqslant r_{\boldsymbol{y}_0}$ the open ball $B_r(\boldsymbol{y}_0)$ is contained in $\mathrm{int}\ (\mathrm{dom}\ J)$. Hence

$$\begin{aligned}
S_\epsilon(\boldsymbol{x},t) &\coloneqq -\epsilon\ln\left(\frac{1}{(2\pi t\epsilon)^{n/2}}\int_{\mathrm{int}\ (\mathrm{dom}\ J)} e^{-(\frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2+J(\boldsymbol{y}))/\epsilon}\, d\boldsymbol{y}\right)\\
&\leqslant -\epsilon\ln\left(\frac{1}{(2\pi t\epsilon)^{n/2}}\int_{B_r(\boldsymbol{y}_0)} e^{-(\frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2+J(\boldsymbol{y}))/\epsilon}\, d\boldsymbol{y}\right)\\
&\leqslant -\epsilon\ln\left(\frac{1}{(2\pi t\epsilon)^{n/2}}\int_{B_r(\boldsymbol{y}_0)} e^{-\frac{1}{2t\epsilon}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2}\, d\boldsymbol{y}\right) + \sup_{\boldsymbol{y}\in B_r(\boldsymbol{y}_0)} J(\boldsymbol{y}).
\end{aligned}$$

Take $\limsup_{\substack{\epsilon\to 0\\ \epsilon>0}}$ and apply the large deviation principle to the term on the right to get

$$\limsup_{\substack{\epsilon\to 0\\ \epsilon>0}} S_\epsilon(\boldsymbol{x},t) \leqslant \operatorname*{ess\,inf}_{\boldsymbol{y}\in B_r(\boldsymbol{y}_0)} \frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + \sup_{\boldsymbol{y}\in B_r(\boldsymbol{y}_0)} J(\boldsymbol{y}).$$

Taking $\lim_{r\to 0}$ on both sides of the inequality yields

$$\limsup_{\substack{\epsilon\to 0\\ \epsilon>0}} S_\epsilon(\boldsymbol{x},t) \leqslant \frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}_0\right\|_2^2 + J(\boldsymbol{y}_0).$$

Since the inequality holds for every $\boldsymbol{y}_0 \in \operatorname{int}(\operatorname{dom} J)$, we can take the infimum over all $y \in \operatorname{int}(\operatorname{dom} J)$ on the right-hand-side of the inequality to get

$$\limsup_{\substack{\epsilon\to 0\\ \epsilon>0}} S_\epsilon(\boldsymbol{x},t) \leqslant \inf_{\boldsymbol{y}\in \operatorname{int}\ (\operatorname{dom}\ J)} \left\{\frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + J(\boldsymbol{y})\right\}.$$

By convexity of J, the infimum on the right hand side is equal to that taken over $\operatorname{dom} J$ ([44], Corollary 7.3.2), i.e.,

$$\inf_{\boldsymbol{y}\in \operatorname{int}\ (\operatorname{dom}\ J)} \left\{\frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + J(\boldsymbol{y})\right\} = \inf_{\boldsymbol{y}\in \operatorname{dom}\ J} \left\{\frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2 + J(\boldsymbol{y})\right\} \equiv S_0(\boldsymbol{x},t).$$

Hence

$$\limsup_{\substack{\epsilon\to 0\\ \epsilon>0}} S_\epsilon(\boldsymbol{x},t) \leqslant S_0(\boldsymbol{x},t).$$

Step 2. We can directly invoke (Deuschel and Stroock [14], Lemma 2.1.8) to get[1]

$$\liminf_{\substack{\epsilon\to 0\\ \epsilon>0}} S_\epsilon(\boldsymbol{x},t) \geqslant S_0(\boldsymbol{x},t).$$

Step 3. Combining the two limits derived in steps 1 and 2 yield

$$\lim_{\substack{\epsilon\to 0\\ \epsilon>0}} S_\epsilon(\boldsymbol{x},t) = S_0(\boldsymbol{x},t)$$

for every $\boldsymbol{x}\in\mathbb{R}^n$ and $t>0$, where convergence is uniform on every closed bounded subset $(\boldsymbol{x},t)$ of $\mathbb{R}^n\times(0,+\infty)$ ([44], Theorem 10.8).

[1] In the notation of Deuschel and Stroock [14], $\Phi = -J$, which is upper semicontinuous, $\boldsymbol{y}\mapsto \frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|_2^2$ is the rate function, and note that the tail condition (2.1.9) is satisfied by assumption (A3) in that $\sup_{\boldsymbol{y}\in\mathbb{R}^n} -J(\boldsymbol{y}) = -\inf_{\boldsymbol{y}\in\mathbb{R}^n} J(\boldsymbol{y}) = 0$.

By differentiability and joint convexity of both $(\boldsymbol{x}, t) \mapsto S_0(\boldsymbol{x}, t)$ and $(\boldsymbol{x}, t) \mapsto S_\epsilon(\boldsymbol{x}, t) - \frac{n\epsilon}{2} \ln t$ (Theorem 2.1 (i), and Theorem 3.1 (i) and (ii)(a)), we can invoke ([44], Theorem 25.7) to get

$$\lim_{\substack{\epsilon \to 0 \\ \epsilon > 0}} \nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x}, t) = \nabla_{\boldsymbol{x}} S_0(\boldsymbol{x}, t) \text{ and } \lim_{\substack{\epsilon \to 0 \\ \epsilon > 0}} \left(\frac{\partial S_\epsilon(\boldsymbol{x}, t)}{\partial t} - \frac{n\epsilon}{2t} \right) = \lim_{\substack{\epsilon \to 0 \\ \epsilon > 0}} \frac{\partial S_\epsilon(\boldsymbol{x}, t)}{\partial t} = \frac{\partial S_0(\boldsymbol{x}, t)}{\partial t},$$

for every $\boldsymbol{x} \in \mathbb{R}^n$ and $t > 0$, where convergence is uniform on every closed bounded subset of $\mathbb{R}^n \times (0, +\infty)$. Furthermore, the viscous HJ PDE (32) for S_ϵ implies that

$$\begin{aligned} \lim_{\substack{\epsilon \to 0 \\ \epsilon > 0}} \frac{\epsilon}{2} \Delta_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x}, t) &= \lim_{\substack{\epsilon \to 0 \\ \epsilon > 0}} - \left(\frac{\partial S_\epsilon(\boldsymbol{x}, t)}{\partial t} + \frac{1}{2} \left\| \nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x}, t) \right\|^2 \right), \\ &= - \left(\frac{\partial S_0(\boldsymbol{x}, t)}{\partial t} + \frac{1}{2} \left\| \nabla_{\boldsymbol{x}} S_0(\boldsymbol{x}, t) \right\|^2 \right) \\ &= 0, \end{aligned}$$

where the last equality holds by the structure of the first-order HJ PDE 22 (see Theorem 2.1). Here, again, the limit holds for every $\boldsymbol{x} \in \mathbb{R}^n$ and $t > 0$, and convergence is uniform over any closed bounded subset of $\mathbb{R}^n \times (0, +\infty)$. Finally, the limit $\lim_{\substack{\epsilon \to 0 \\ \epsilon > 0}} \boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) = \boldsymbol{u}_{MAP}(\boldsymbol{x}, t)$ holds directly as a consequence to the limit $\lim_{\substack{\epsilon \to 0 \\ \epsilon > 0}} \nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x}, t) = \nabla_{\boldsymbol{x}} S_0(\boldsymbol{x}, t)$ and the representation formulas (34) and (26) for the posterior mean and MAP estimates, respectively.

Appendix B. Proof of Proposition 4.1

We will prove that $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) \in \text{int}\ (\text{dom}\ J)$ in two steps. First, we will use the linearity of the projection operator (12) and the posterior mean estimate to prove by contradiction that $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) \in \text{cl}\ (\text{dom}\ J)$. Second, we will use the following variant of the Hahn–Banach theorem for convex body in $\mathbb{R}^n$ to show in fact that $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) \in \text{int}\ (\text{dom}\ J)$.

Theorem B.1. *([44], Theorem 11.6 and Corollary 11.6.2) Let C be a convex set. A point $\boldsymbol{x} \in C$ is a relative boundary point of C if and only if there exist a vector $\boldsymbol{a} \in \mathbb{R}^n \backslash \{\boldsymbol{0}\}$ and a number $b \in \mathbb{R}$ such that*

$$\boldsymbol{x} = \arg\max_{\boldsymbol{y} \in C} \langle \boldsymbol{a}, \boldsymbol{y} \rangle + b,$$

i.e., there exists an affine function on C that is not identically constant and achieves its maximum over C at $\boldsymbol{x}$.

Step 1. Suppose $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon) \notin \text{cl}\ (\text{dom}\ J)$. Since the set cl (dom J) is closed and convex, the projection of $\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)$ onto cl (dom J) given by $\pi_{\text{cl}(\text{dom}\ J)}(\boldsymbol{u}_{PM}(\boldsymbol{x}, t, \epsilon)) \equiv \bar{\boldsymbol{u}}$ is well-defined and unique (see Definition 4), with $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) \neq \bar{\boldsymbol{u}}$ by assumption. It also satisfies the characterization (13), namely

$$\langle \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) - \bar{\boldsymbol{u}}, \boldsymbol{y} - \bar{\boldsymbol{u}})\rangle \leqslant 0$$

for every $\boldsymbol{y} \in \mathrm{cl}\ (\mathrm{dom}\ J)$. However, by linearity of the posterior mean estimate,

$$\|\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) - \bar{\boldsymbol{u}}\|^2 = \mathbb{E}_J\left[\langle \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) - \bar{\boldsymbol{u}}, \boldsymbol{y} - \bar{\boldsymbol{u}}\rangle)\right] \leqslant 0,$$

which implies that $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) = \bar{\boldsymbol{u}}$, in contradiction with the assumption that $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) \notin \mathrm{cl}\ (\mathrm{dom}\ J)$.

Step 2. Suppose $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) \notin \mathrm{int}\ (\mathrm{dom}\ J)$. We know $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) \in \mathrm{cl}\ (\mathrm{dom}\ J)$ by the first step, and therefore $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)$ must be a boundary point of dom J. If J has no boundary point, then we get a contradiction and conclude $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) \in \mathrm{int}\ (\mathrm{dom}\ J)$. If not, Theorem B.1 applies and therefore there exist a vector $\boldsymbol{a} \in \mathbb{R}^n \backslash \{\boldsymbol{0}\}$ and a number $b \in \mathbb{R}$ such that $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) = \arg\max_{\boldsymbol{y} \in \mathrm{cl}\ (\mathrm{dom}\ J)} \{\langle \boldsymbol{a}, \boldsymbol{y}\rangle + b\}$, with $\langle \boldsymbol{a}, \boldsymbol{y}\rangle + b < \langle \boldsymbol{a}, \boldsymbol{u}_{PM}(\boldsymbol{x},t)\rangle + b$ for every $\boldsymbol{y} \in \mathrm{int}\ (\mathrm{dom}\ J)$. However, by linearity of the posterior mean estimate,

$$\begin{aligned}
\langle \boldsymbol{a}, \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)\rangle + b &= \mathbb{E}_J\left[\langle \boldsymbol{a}, \boldsymbol{y}\rangle + b\right] \\
&< \mathbb{E}_J\left[\langle \boldsymbol{a}, \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)\rangle + b\right] \\
&= \langle \boldsymbol{a}, \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)\rangle + b,
\end{aligned}$$

where the strict inequality in the second line follows from integrating over int (dom J). This gives a contradiction, and hence $\boldsymbol{u}_{PM}(\boldsymbol{x},t) \in \mathrm{int}\ (\mathrm{dom}\ J)$.

As a consequence, the subdifferential of J at $\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)$ is non-empty because the subdifferential ∂J is non-empty at every point $\boldsymbol{y} \in \mathrm{int}\ (\mathrm{dom}\ J)$ ([44], Theorem 23.4). Hence there exists a subgradient $\boldsymbol{p} \in \partial J(u_{PM}(\boldsymbol{x},t,\epsilon))$ such that

$$J(\boldsymbol{y}) \geqslant J(\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)) - \langle \boldsymbol{p}, \boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)\rangle\,.$$

Taking the expectation $\mathbb{E}_J[\cdot]$ on both sides yield the inequality $J(\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)) \leqslant \mathbb{E}_J[J(\boldsymbol{y})]$. Now, the convex inequality $1 + z \leqslant e^z$ (which holds for every $z \in \mathbb{R} \cup \{+\infty\}$ with the understanding that $+\infty \leqslant +\infty$) with the choice of $z = J(\boldsymbol{y})/\epsilon$ for $\boldsymbol{y} \in \mathbb{R}^n$ yield $J(\boldsymbol{y})e^{-J(\boldsymbol{y})/\epsilon} \leqslant \epsilon(1 - e^{-J(\boldsymbol{y})/\epsilon})$.

After multiplying both sides by $\frac{1}{Z_J(\boldsymbol{x},t,\epsilon)}e^{-\frac{1}{2t\epsilon}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2}$ and integrating with respect to $\boldsymbol{y}$, we find $\mathbb{E}_J\left[J(\boldsymbol{y})\right] \leqslant \epsilon\left(e^{S_\epsilon(\boldsymbol{x},t)/\epsilon}-1\right) < +\infty$.

Appendix C. Proof of Proposition 4.2

Proof of (i): Here, we suppose dom $J = \mathbb{R}^n$. Let D_J denote the set of points at which J is continuously differentiable on $\mathbb{R}^n$, let $\nabla J(\boldsymbol{y})$ denote the gradient of J at these points, let N_J denote the set of the points at which J is not continuously differentiable on $\mathbb{R}^n$, and let $\boldsymbol{v} \in \mathbb{R}^n \setminus \{\boldsymbol{0}\}$ by any nonzero vector in $\mathbb{R}^n$. We will derive the representation formulas (41) and (42) by showing that the divergence of the vector fields $\boldsymbol{y} \mapsto \boldsymbol{y}e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon}$ and $\boldsymbol{y} \mapsto \boldsymbol{v}e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon}$ integrate to zero on D_J, i.e.,

$$\int_{D_J} \nabla_{\boldsymbol{y}} \cdot \left(\boldsymbol{v}e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)\epsilon}\right) d\boldsymbol{y} = 0. \tag{55}$$

and

$$\int_{D_J} \nabla_{\boldsymbol{y}} \cdot \left(\boldsymbol{y}e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)\epsilon}\right) d\boldsymbol{y} = 0. \tag{56}$$

We will first assume that these equations hold and derive the representation formulas (41) and (42), and we will then prove that equations (55) and (56) hold.

Suppose that equations (55) and (56) hold. First, write the integral in (55) as

$$\left\langle \boldsymbol{v}, \int_{D_J} \left(\frac{\boldsymbol{y}-\boldsymbol{x}}{t} + \nabla J(\boldsymbol{y})\right) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y}\right\rangle = 0.$$

As $\boldsymbol{v}$ is an arbitrary nonzero vector, the integral in the inner product is equal to zero. Since the minimal subgradient $\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0}) = \nabla J(\boldsymbol{y})$ everywhere on D_J, the set D_J is dense in $\mathbb{R}^n$, and the n-dimensional Lebesgue measure of N_J is zero ([44], Theorem 25.5), the gradient $\nabla J(\boldsymbol{y})$ in this integral may be replaced with the minimal subgradient $\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})$ and the domain be taken to be $\mathbb{R}^n$ without changing the value of the integral. Hence, we have

$$\int_{\mathbb{R}^n} \left(\frac{\boldsymbol{y}-\boldsymbol{x}}{t} + \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y} = 0.$$

Dividing through by the partition function $Z_J(\boldsymbol{x},t,\epsilon)$ and rearranging yield

$$\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) = \boldsymbol{x} - t\mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right],$$

which is the representation formula (41). In particular, we find the representation formula $\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t) = \mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right]$ via Equation (34) in Theorem 3.1(iii). Next, write the integral in (56) as

$$\int_{D_J}\left(n\epsilon - \left\langle \boldsymbol{y}, \left(\frac{\boldsymbol{y}-\boldsymbol{x}}{t} + \nabla J(\boldsymbol{y})\right)\right\rangle\right) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{y} = 0.$$

As discussed previously, we may replace the gradient $\nabla J(\boldsymbol{y})$ with the minimal subgradient $\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})$ and take the domain to be $\mathbb{R}^n$ in this integral without affecting its value. With these changes and on dividing through by the partition function $Z_J(\boldsymbol{x},t,\epsilon)$, we find

$$\mathbb{E}_J\left[\left\langle \boldsymbol{y}, \left(\frac{\boldsymbol{y}-\boldsymbol{x}}{t} + \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right)\right\rangle\right] = n\epsilon.$$

We can re-write this as

$$\mathbb{E}_J\left[\langle \boldsymbol{y}, \boldsymbol{y}-\boldsymbol{x}\rangle + t\left\langle \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon), \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right\rangle\right] = nt\epsilon - t\mathbb{E}_J\left[\left\langle \boldsymbol{y} - \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon), \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right\rangle\right],$$

and we can write the left hand side as $\mathbb{E}_J\left[\|\boldsymbol{y}-\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)\|_2^2\right]$ using the representation formula $\mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right] = \left(\frac{\boldsymbol{x}-\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)}{t}\right)$ derived previously, which gives the representation formula (42). In particular, we find the representation formula $\Delta_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t) = \frac{1}{t\epsilon}\mathbb{E}_J\left[\left\langle \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0}), \boldsymbol{y}-\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)\right\rangle\right]$ via Equation (35) in Theorem 3.1(iii).

We now establish the two equalities (55) and (56). To do so, we will use a measure-theoretic version of the divergence theorem due to Pfeffer [41] that will apply to the vector fields $\boldsymbol{y}e^{-\left(\frac{1}{2t\epsilon}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)}$ and $\boldsymbol{v}e^{-\left(\frac{1}{2t\epsilon}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)}$.

Let $r > 0$ and $B_r(\boldsymbol{0}) \coloneqq \{\boldsymbol{y} \in \mathbb{R}^n\colon\ \|\boldsymbol{y}\|_2 \leqslant r\}$ denote the set of points in the open ball of radius r centered at $\boldsymbol{0}$ in $\mathbb{R}^n$. As $B_r(\boldsymbol{0})$ is a bounded, convex, and open set on which the function J is Lipschitz continuous ([44], Theorem 10.4), the set of points N_J at which J is not differentiable in the closed ball cl $B_r(\boldsymbol{0})$ constitutes a σ-finite measurable set with respect to the $n-1$ dimensional Lebesgue measure ([1], Theorem 4.1). Thanks to these properties, we can invoke ([41], Theorem 4.14) to conclude that there exist two unit normal vectors $\boldsymbol{n}_{\boldsymbol{y}}$ and $\boldsymbol{n}_{\boldsymbol{v}}$ such that

$$\int_{D_J \cap \text{ cl } B_r(\boldsymbol{0})} \nabla_{\boldsymbol{y}} \cdot \left(\boldsymbol{v}e^{-(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y}))/\epsilon}\right) d\boldsymbol{y} = \int_{\text{bd } (D_J \cap \text{ cl } B_r(\boldsymbol{0}))} (\boldsymbol{v}\cdot\boldsymbol{n}_{\boldsymbol{v}}) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{S}$$

and

$$\int_{D_J \cap \text{ cl } B_r(\boldsymbol{0})} \nabla_{\boldsymbol{y}} \cdot \left(\boldsymbol{y}e^{-(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y}))/\epsilon}\right) d\boldsymbol{y} = \int_{\text{bd } (D_J \cap \text{ cl } B_r(\boldsymbol{0}))} (\boldsymbol{y}\cdot\boldsymbol{n}_{\boldsymbol{y}}) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}\, d\boldsymbol{S},$$

where dS denotes the $n-1$ dimensional Lebesgue measure. Take the limit $r \to +\infty$ on both sides to get

$$\int_{D_J} \nabla_{\boldsymbol{y}} \cdot \left(\boldsymbol{v} e^{-(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y}))/\epsilon}\right) d\boldsymbol{y} = \lim_{r\to+\infty} \int_{\text{bd } (D_J \cap \text{ cl } B_r(\boldsymbol{0}))} (\boldsymbol{v} \cdot \boldsymbol{n_v}) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon} \, d\boldsymbol{S}$$

and

$$\int_{D_J} \nabla_{\boldsymbol{y}} \cdot \left(\boldsymbol{y} e^{-(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y}))/\epsilon}\right) d\boldsymbol{y} = \lim_{r\to+\infty} \int_{\text{bd } (D_J \cap \text{ cl } B_r(\boldsymbol{0}))} (\boldsymbol{y} \cdot \boldsymbol{n_y}) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon} \, d\boldsymbol{S},$$

The right hand side of these equations are equal to zero since the limits

$$0 \leqslant \lim_{\|\boldsymbol{y}\|_2\to+\infty} \left\|(\boldsymbol{v} \cdot \boldsymbol{n_v}) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}\right\|_2 \leqslant \lim_{\|\boldsymbol{y}\|_2\to+\infty} \|\boldsymbol{v}\|_2 \, e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2\right)/\epsilon} = 0$$

and

$$0 \leqslant \lim_{\|\boldsymbol{y}\|_2\to+\infty} \left\|(\boldsymbol{y} \cdot \boldsymbol{n_y}) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}\right\|_2 \leqslant \lim_{\|\boldsymbol{y}\|_2\to+\infty} \|\boldsymbol{y}\|_2 \, e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2\right)/\epsilon} = 0$$

and a short calculation yield

$$\lim_{r\to+\infty} \int_{\text{bd } (D_J \cap \text{ cl } B_r(\boldsymbol{0}))} (\boldsymbol{v} \cdot \boldsymbol{n_v}) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon}) \, d\boldsymbol{S} = 0$$

and

$$\lim_{r\to+\infty} \int_{\text{bd } (D_J \cap \text{ cl } B_r(\boldsymbol{0}))} (\boldsymbol{v} \cdot \boldsymbol{n_v}) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right)/\epsilon} \, d\boldsymbol{S}.$$

Hence Equations (55) and (56), and the proof of (i) is finished.

Proof of (ii): Let $\{\mu_k\}_{k=1}^{+\infty}$ be a sequence of positive real numbers converging to zero and $\boldsymbol{y} \in \text{dom } J$. By Theorem 2.1(i), the sequence of real numbers $\{S_0(\boldsymbol{x}, \mu_k)\}_{k=1}^{+\infty}$ converges to $J(\boldsymbol{y})$, and by assumption (A3) that $\inf_{\boldsymbol{y}\in\mathbb{R}^n} J(\boldsymbol{y}) = 0$ the sequence $\{S_0(\boldsymbol{x}, \mu_k)\}_{k=1}^{+\infty}$ is bounded uniformly from below by 0, i.e.,

$$\begin{aligned} S_0(\boldsymbol{x}, \mu_k) &= \inf_{\boldsymbol{y}\in\mathbb{R}^n} \left\{\frac{1}{2\mu_k} \|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})\right\} \\ &\geqslant \inf_{\boldsymbol{y}\in\mathbb{R}^n} \left\{\frac{1}{2\mu_k} \|\boldsymbol{x}-\boldsymbol{y}\|_2^2\right\} + \inf_{\boldsymbol{y}\in\mathbb{R}^n} J(\boldsymbol{y}) \\ &= 0, \end{aligned}$$

and therefore,

$$\frac{1}{(2\pi t\epsilon)^{n/2}} \int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + S_0(\boldsymbol{x},\mu_k)\right)/\epsilon} \, d\boldsymbol{y} \leqslant \frac{1}{(2\pi t\epsilon)^{n/2}} \int_{\mathbb{R}^n} e^{-\frac{1}{2t\epsilon}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2} \, d\boldsymbol{y} = 1.$$

Using the Lebesgue dominated convergence theorem ([18], Theorem 2.24) and the limit result $\lim_{k\to+\infty} e^{-S_0(\boldsymbol{x},\mu_k)/\epsilon} = e^{-J(\boldsymbol{y})/\epsilon}$ for every $\boldsymbol{y} \in \mathrm{dom}\ J$ from Theorem 2.1(i) (with $\lim_{k\to+\infty} e^{-S_0(\boldsymbol{x},\mu_k)/\epsilon} = 0$ for every $\boldsymbol{y} \notin \mathrm{dom}\ J$), we find

$$\lim_{k\to+\infty} -\epsilon \log\left(\frac{1}{(2\pi t\epsilon)^{n/2}}\int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+S_0(\boldsymbol{x},\mu_k)\right)/\epsilon}\,d\boldsymbol{y}\right) = S_\epsilon(\boldsymbol{x},t).$$

Thanks to the convexity of $\boldsymbol{x} \mapsto S_\epsilon(\boldsymbol{x},t)$ proven in Theorem 3.1(ii)(a) as well as the representation formula 41 derived in part (i) of this proof, we can invoke ([44], Theorem 25.7) to find

$$\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x},t) = \lim_{k\to+\infty}\left(\frac{\int_{\mathbb{R}^n} \nabla_{\boldsymbol{y}} S_0(\boldsymbol{y},\mu_k) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+S_0(\boldsymbol{y},\mu_k)\right)/\epsilon}\,d\boldsymbol{y}}{\int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+S_0(\boldsymbol{y},\mu_k)\right)/\epsilon}\,d\boldsymbol{y}}\right).$$

Finally, we can use formula (34) and the limit above to find

$$\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) = \boldsymbol{x} - t\lim_{k\to+\infty}\left(\frac{\int_{\mathbb{R}^n} \nabla_{\boldsymbol{y}} S_0(\boldsymbol{y},\mu_k) e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+S_0(\boldsymbol{y},\mu_k)\right)/\epsilon}\,d\boldsymbol{y}}{\int_{\mathbb{R}^n} e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+S_0(\boldsymbol{y},\mu_k)\right)/\epsilon}\,d\boldsymbol{y}}\right).$$

Appendix D. Proof of Proposition 4.3

Let $\{\mu_k\}_{k=1}^{+\infty}$ be a sequence of positive real numbers converging to zero and let $S_0 \colon \mathbb{R}^n \times (0,+\infty) \mapsto \mathbb{R}$ denote the solution to the first-order HJ PDE (22) with initial data J. Define the function $F \colon \mathrm{dom}\ \partial J \times \mathrm{dom}\ \partial J \times \mathbb{R}^n \times (0,+\infty) \mapsto [0,+\infty)$ as

$$F(\boldsymbol{y},\boldsymbol{y}_0,\boldsymbol{x},t) = \left\langle \left(\frac{\boldsymbol{y}-\boldsymbol{x}}{t} + \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right) - \left(\frac{\boldsymbol{y}_0-\boldsymbol{x}}{t} + \pi_{\partial J(\boldsymbol{y}_0)}(\boldsymbol{0})\right), \boldsymbol{y}-\boldsymbol{y}_0\right\rangle e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y})\right)/\epsilon}$$

and the sequence of functions $\{F_{\mu_k}\}_{k=1}^{+\infty}$ with $F_{\mu_k} \colon \mathrm{dom}\ \partial J \times \mathrm{dom}\ \partial J \times \mathbb{R}^n \times (0,+\infty) \mapsto [0,+\infty)$ as

$$F_{\mu_k}(\boldsymbol{y},\boldsymbol{y}_0,\boldsymbol{x},t) = \left\langle \left(\frac{\boldsymbol{y}-\boldsymbol{x}}{t} + \nabla_{\boldsymbol{y}} S_0(\boldsymbol{y},\mu_k)\right) - \left(\frac{\boldsymbol{y}_0-\boldsymbol{x}}{t} + \nabla_{\boldsymbol{y}} S_0(\boldsymbol{y}_0,\mu_k)\right), \boldsymbol{y}-\boldsymbol{y}_0\right\rangle e^{-\left(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+S_0(\boldsymbol{y},\mu_k)\right)/\epsilon}.$$

Since $\lim_{k\to+\infty} S_0(\boldsymbol{y},\mu_k) = J(\boldsymbol{y})$ and $\lim_{k\to+\infty} \nabla_{\boldsymbol{y}} S_0(\boldsymbol{y},\mu_k) = \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})$ for every $\boldsymbol{y} \in \mathbb{R}^n$ by Theorem 2.1(i) and (iv), the limit $\lim_{k\to+\infty} F_\mu(\boldsymbol{y},\boldsymbol{y}_0,\boldsymbol{x},t) = F(\boldsymbol{y},\boldsymbol{y}_0,\boldsymbol{x},t)$ holds for every $\boldsymbol{y} \in \mathrm{dom}\ \partial J$, $\boldsymbol{y}_0 \in \mathrm{dom}\ \partial J$, $\boldsymbol{x} \in \mathbb{R}^n$ and $t > 0$. Moreover, the function F and sequence of functions $\{F_{\mu_k}\}_{k=1}^{+\infty}$ are positive functions in their arguments because the strong convexity of both $\mathrm{dom}\ \partial J \ni \boldsymbol{y} \mapsto \frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + J(\boldsymbol{y})$ and $\mathrm{dom}\ \partial J \ni \boldsymbol{y} \mapsto \frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 + S_0(\boldsymbol{y},\mu)$ with modulus $\left(\frac{1+mt}{t}\right)$ imply

$$\left(\frac{1+mt}{t}\right)\|\boldsymbol{y}-\boldsymbol{y}_0\|_2^2 \leqslant \left\langle \left(\frac{\boldsymbol{y}-\boldsymbol{x}}{t} + \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})\right) - \left(\frac{\boldsymbol{y}_0-\boldsymbol{x}}{t} + \pi_{\partial J(\boldsymbol{y}_0)}(\boldsymbol{0})\right), \boldsymbol{y}-\boldsymbol{y}_0\right\rangle \tag{57}$$

and

$$\left(\frac{1+mt}{t}\right)\|\boldsymbol{y}-\boldsymbol{y}_0\|_2^2 \leqslant \left\langle \left(\frac{\boldsymbol{y}-\boldsymbol{x}}{t}+\nabla_{\boldsymbol{y}} S_0(\boldsymbol{y},\mu_k)\right)-\left(\frac{\boldsymbol{y}_0-\boldsymbol{x}}{t}+\nabla_{\boldsymbol{y}} S_0(\boldsymbol{y}_0,\mu_k)\right), \boldsymbol{y}-\boldsymbol{y}_0\right\rangle.$$

As a consequence, Fatou's lemma applies to the sequence of functions $\{F_{\mu_k}\}_{k=1}^{+\infty}$, and hence

$$\begin{aligned}\int_{\mathrm{dom}\ \partial J} F(\boldsymbol{y},\boldsymbol{y}_0,\boldsymbol{x},t)\,d\boldsymbol{y} &\leqslant \liminf_{k\to+\infty}\int_{\mathrm{dom}\ \partial J} F_{\mu_k}(\boldsymbol{y},\boldsymbol{y}_0,\boldsymbol{x},t)\,d\boldsymbol{y}\\ &\leqslant \liminf_{k\to+\infty}\int_{\mathbb{R}^n} F_{\mu_k}(\boldsymbol{y},\boldsymbol{y}_0,\boldsymbol{x},t)\,d\boldsymbol{y}.\end{aligned}$$

By the representation formulas (41) and (42) derived in Proposition 4.2 applied to the initial data $\boldsymbol{y}\mapsto S_0(\boldsymbol{y},\mu)$,

$$\int_{\mathbb{R}^n} F_{\mu_k}(\boldsymbol{y},\boldsymbol{y}_0\boldsymbol{x},t)\,d\boldsymbol{y} = \int_{\mathbb{R}^n}\left(n\epsilon - \left\langle \frac{\boldsymbol{y}_0-\boldsymbol{x}}{t}+\nabla_{\boldsymbol{y}} S_0(\boldsymbol{y}_0,\mu_k), \boldsymbol{y}-\boldsymbol{y}_0\right\rangle\right) e^{-(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+S_0(\boldsymbol{y},\mu_k))/\epsilon}\,d\boldsymbol{y}.$$

A straightforward calculation using assumption (A3) yields $S_0(\boldsymbol{y},\mu_k)\geqslant 0$ for every $k\in\mathbb{N}$, and it implies $\|\boldsymbol{y}-\boldsymbol{y}_0\|_2\, e^{-(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+S_0(\boldsymbol{y},\mu_k))/\epsilon} \leqslant \|\boldsymbol{y}-\boldsymbol{y}_0\|_2\, e^{-(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2)/\epsilon}$, which is integrable over $\mathbb{R}^n$. Hence the Lebesgue dominated convergence theorem applies and since $\lim_{k\to+\infty}\nabla_{\boldsymbol{y}} S_0(\boldsymbol{y},\mu_k) = \pi_{\partial J(\boldsymbol{y})}(\boldsymbol{0})$ by Theorem 2.1(iv), we get

$$\liminf_{k\to+\infty}\int_{\mathbb{R}^n} F_{\mu_k}(\boldsymbol{y},\boldsymbol{x},t)\,d\boldsymbol{y} = n\epsilon - \left\langle \frac{\boldsymbol{y}_0-\boldsymbol{x}}{t}+\pi_{\partial J(\boldsymbol{y}_0)}(\boldsymbol{0}), \int_{\mathrm{dom}\ \partial J}(\boldsymbol{y}-\boldsymbol{y}_0)e^{-(\frac{1}{2t}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2+J(\boldsymbol{y}))/\epsilon}\,d\boldsymbol{y}\right\rangle.$$

Finally, combining this limit with the strong convexity inequality (57) and dividing through by the partition function $Z_J(\boldsymbol{x},t,\epsilon)$, we get inequality (44), which proves Proposition 4.3.

Appendix E. Proof of Theorem 4.1

Proof of (i): For every $\boldsymbol{u}\in\mathrm{dom}\ \partial J$, the Bregman divergence of the function $\mathbb{R}^n\ni\boldsymbol{y}\mapsto\Phi_J(\boldsymbol{y},\boldsymbol{x},t)$ at $(\boldsymbol{y},\varphi_J(\boldsymbol{u},\boldsymbol{x},t))$ is given by

$$\begin{aligned}D_{\Phi_J}(\boldsymbol{y},\varphi_J(\boldsymbol{u},\boldsymbol{x},t)) &= \Phi_J(\boldsymbol{y},\boldsymbol{x},t)-\langle\varphi_J(\boldsymbol{u},\boldsymbol{x},t),\boldsymbol{y}\rangle+\Phi_J^*(\varphi_J(\boldsymbol{u},\boldsymbol{x},t),x,t),\\ &= \Phi_J(\boldsymbol{y},\boldsymbol{x},t)-\Phi_J(\boldsymbol{u},\boldsymbol{x},t)-\langle\varphi_J(\boldsymbol{u},\boldsymbol{x},t),\boldsymbol{y}-\boldsymbol{u}\rangle\end{aligned}$$

by definition of the subdifferential (see definition 6), where equality holds because $\varphi_J(\boldsymbol{u},\boldsymbol{x},t)\in\partial\Phi_J(\boldsymbol{u},\boldsymbol{x},t)$. Note that the expected value $\mathbb{E}_J[\Phi_J(\boldsymbol{y},\boldsymbol{x},t)]$ is finite because $\mathbb{E}_J[J(\boldsymbol{y})]$ is finite by assumption (A3) and Proposition 4.1. Hence

$$\begin{aligned}\mathbb{E}_J\left[D_{\Phi_J}(\boldsymbol{y},\varphi_J(\boldsymbol{u},\boldsymbol{x},t)) - D_{\Phi_J}(\boldsymbol{y},\varphi_J(\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon),\boldsymbol{x},t)\right] &= \Phi_J(\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon),\boldsymbol{x},t) - \Phi_J(\boldsymbol{u},\boldsymbol{x},t)\\ &\quad - \left\langle \varphi_J(\boldsymbol{u},\boldsymbol{x},t), \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon) - \boldsymbol{u}\right\rangle,\\ &\geqslant \frac{1}{t}\left\|\boldsymbol{u}_{PM} - \boldsymbol{u}\right\|_2^2.\end{aligned}$$

where the inequality holds because $\varphi_J(\boldsymbol{u},\boldsymbol{x},t)$ is a subgradient of $\Phi_J(\boldsymbol{u},\boldsymbol{x},t)$ and $\boldsymbol{y}\mapsto\Phi_J(\boldsymbol{y},\boldsymbol{x},t)$ is strongly convex of parameter $\frac{1}{t}$. This shows $\mathbb{E}_J\left[D_{\Phi_J}(\boldsymbol{y},\varphi_J(\boldsymbol{u},\boldsymbol{x},t))\right] \geqslant \mathbb{E}_J\left[D_{\Phi_J}(\boldsymbol{y},\varphi_J(\boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon),\boldsymbol{x},t)\right]$ for every $\boldsymbol{u}\in\operatorname{dom}\partial J$, with equality if and only if $\boldsymbol{u} = \boldsymbol{u}_{PM}(\boldsymbol{x},t,\epsilon)$.

Proof of (ii): For every $\boldsymbol{u}\in\mathbb{R}^n$, the Bregman divergence of $\operatorname{dom}\partial J \ni \boldsymbol{y}\mapsto\Phi_J(\boldsymbol{y},\boldsymbol{x},t)$ at $(\boldsymbol{u},\varphi_J(\boldsymbol{y},\boldsymbol{x},t))$ is given by

$$\begin{aligned}D_{\Phi_J}(\boldsymbol{u},\varphi_J(\boldsymbol{y},\boldsymbol{x},t)) &= \Phi_J(\boldsymbol{u},\boldsymbol{x},t) - \left\langle\varphi_J(\boldsymbol{y},\boldsymbol{x},t),\boldsymbol{u}\right\rangle + \Phi_J^*(\varphi_J(\boldsymbol{y},\boldsymbol{x},t))\\ &= \Phi_J(\boldsymbol{u},\boldsymbol{x},t) - \Phi_J(\boldsymbol{y},\boldsymbol{x},t) - \left\langle\varphi_J(\boldsymbol{y},\boldsymbol{x},t),\boldsymbol{u}-\boldsymbol{y}\right\rangle\end{aligned}$$

by definition of the subdifferential (see definition 6), where equality holds because $\varphi_J(\boldsymbol{y},\boldsymbol{x},t)\in\partial\Phi_J(\boldsymbol{y},\boldsymbol{x},t)$. Note that the expected value $\mathbb{E}_J\left[D_{\Phi_J}(\boldsymbol{u},\varphi_J(\boldsymbol{y},\boldsymbol{x},t))\right]$ is finite because the expected value $\mathbb{E}_J\left[\left\langle\varphi_J(\boldsymbol{y},\boldsymbol{x},t),\boldsymbol{u}-\boldsymbol{y}\right\rangle\right]$ in the second line of the previous equation is finite thanks to the monotonicity property (44) and finiteness of the mean minimal subgradient $\mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\mathbf{0})\right]$ by Corollary 4.1. Now, we have

$$\begin{aligned}\mathbb{E}_J\left[D_{\Phi_J}(\boldsymbol{u},\varphi_J(\boldsymbol{y},\boldsymbol{x},t))\right] &= \Phi_J(\boldsymbol{u},\boldsymbol{x},t) - \left\langle\mathbb{E}_J\left[\varphi_J(\boldsymbol{y},\boldsymbol{x},t)\right],\boldsymbol{u}\right\rangle + \mathbb{E}_J\left[\Phi_J^*(\varphi_J(\boldsymbol{y},\boldsymbol{x},t))\right]\\ &= \Phi_J(\boldsymbol{u},\boldsymbol{x},t) + \left\langle\nabla_{\boldsymbol{x}}S_\epsilon(\boldsymbol{x},t) - \mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\mathbf{0})\right],\boldsymbol{u}\right\rangle + \mathbb{E}_J\left[\Phi_J^*(\varphi_J(\boldsymbol{y},\boldsymbol{x},t))\right]\\ &= \frac{1}{2t}\left\|\boldsymbol{x}-\boldsymbol{u}\right\|_2^2 + \left(J(\boldsymbol{u}) + \left\langle\nabla_{\boldsymbol{x}}S_\epsilon(\boldsymbol{x},t) - \mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\mathbf{0})\right],\boldsymbol{u}\right\rangle\right) + \mathbb{E}_J\left[\Phi_J^*(\varphi_J(\boldsymbol{y},\boldsymbol{x},t))\right].\end{aligned}$$

Since $\boldsymbol{u}\mapsto J(\boldsymbol{u}) + \left\langle\nabla_{\boldsymbol{x}}S_\epsilon(\boldsymbol{x},t) - \mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\mathbf{0})\right],\boldsymbol{u}\right\rangle$ is a convex function and $J\in\Gamma_0(\mathbb{R}^n)$ by assumption (A1), we can invoke Theorem 2.1(ii) to conclude that $\boldsymbol{u}\mapsto\mathbb{E}_J\left[D_{\Phi_J}(\boldsymbol{u},\varphi_J(\boldsymbol{y},\boldsymbol{x},t))\right]$ has a unique minimizer $\bar{\boldsymbol{u}}$ that satisfies the inclusion

$$\left(\frac{\boldsymbol{x}-\bar{\boldsymbol{u}}}{t}\right) \in \partial J(\bar{\boldsymbol{u}}) + \left(\nabla_{\boldsymbol{x}}S_\epsilon(\boldsymbol{x},t) - \mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\mathbf{0})\right]\right).$$

.

Proof of (iii): If $\operatorname{dom} J = \mathbb{R}^n$, then the representation formula $\nabla_{\boldsymbol{x}} S_\epsilon(\boldsymbol{x}, t) = \mathbb{E}_J\left[\pi_{\partial J(\boldsymbol{y})}(\mathbf{0})\right]$ derived in Proposition 4.2 holds and the characterization of the minimizer $\bar{\boldsymbol{u}}$ in equation (51) reduces to $\left(\frac{\boldsymbol{x}-\bar{\boldsymbol{u}}}{t}\right) \in \partial J(\bar{\boldsymbol{u}})$. The unique minimizer that satisfies this characterization is the MAP estimate $\boldsymbol{u}_{MAP}(\boldsymbol{x}, t)$ (Theorem 2.1(ii)), i.e., $\bar{\boldsymbol{u}} = \boldsymbol{u}_{MAP}(\boldsymbol{x}, t)$.

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